A New Method of Solving PDEs

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Abstract

The author has developed an algorithm for finding exact, non-separable solutions of partial differential equations and of systems of them. The algorithm uses differential Thomas decompositon, Ritt's full reduction algorithm [1] and the GTZ algorithm [2] to find ODEs that solve the system. We prove a completeness theorem for the core algorithm, and demonstrate how to use the algorithm to solve hydrogen's Schrödinger equation.

Introduction

Starting with the following Schrödinger equation for hydrogen [3]:

$$-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi \tag{1}$$

We can require the solution to solve the following parameterized ODE:

$$(a_0 + a_1 v)\frac{d^2\Psi(v)}{dv^2} + (b_0 + b_1 v)\frac{d\Psi(v)}{dv} + (c_0 + c_1 v)\Psi(v) = 0 \tag{2}$$

where $\Psi(v)$ is a univariate function of some variable v, itself expressed in a parameterized form:

$$v = v_1 x + v_2 y + v_3 z + v_4 r \tag{3}$$

and expect to find that for certain constant values:

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$$\begin{gathered} a_1 = v_4 = 1 \qquad b_0 = c_0 = 2 \qquad c_1 = E \\ a_0 = b_1 = v_1 = v_2 = v_3 = 0 \end{gathered} \tag{4}$$

we will obtain the classical radial equation for hydrogen:

$$r\frac{d^2\Psi(r)}{dr^2} + 2\frac{d\Psi(r)}{dr} + 2(1 + Er)\Psi(r) = 0 \tag{5}$$

a result more commonly derived using separation of variables. Solving the radial equation yields the ground state solution of hydrogen:

$$\begin{gathered} \Psi(r) = e^{-r} \\ E = -1/2 \end{gathered} \tag{6}$$

What might not be expected is to find a different set of constant values:

$$\begin{gathered} E = 0 \\ a_0 = 0 \qquad a_1 = 1 \\ b_0 = -1 \qquad b_1 = 0 \\ c_0 = -1 \qquad c_1 = 0 \\ v_1 = 1 \qquad v_2 = 0 \qquad v_3 = 0 \qquad v_4 = 1 \end{gathered} \tag{7}$$

that produce a different ODE:

$$v\frac{d^2\Psi}{dv}(v) + \frac{d\Psi}{dv}(v) + \Psi(v) = 0 \tag{8}$$

which can be solved to find a different solution to (1):

$$\Psi = J_0(2\sqrt{x + r}) \tag{9}$$

where x, y, z are Cartesian coordinates, $r = \sqrt{x^2 + y^2 + z^2}$, $E = 0$, and J_0 is the ordinary Bessel function J_0. This solution is not of the familiar spherical separated form $R(r)\Theta(\theta)\Phi(\varphi)$, nor is it Cartesian-separable; it is, however, separable in parabolic coordinates, a reflection of the hydrogen atom's hidden symmetry that we take up in §3.

The author has developed an algorithm that tests a system of differential equations, specified as a finite set of differential polynomials, against a second such set (which I term an "ansatz"), parameterized by a finite number of constants, similar to (2) and (3), and finds solutions like (4) and (7). This is done using the techniques of differential algebra and algebraic geometry. The restriction to differential polynomials is not severe; most elementary and special functions can be easily modeled, as will be shown below. Nothing in the development requires the system to consist of partial differential equations, nor to contain more than one of them: a single PDE is the case treated in §3, a system of four in §4.

In short, to solve the system, we restrict our attention to a subset of the function space, parameterized by a finite number of constants and described using differential polynomials. A quick glance at Section 1 should indicate the great variety of such "ansätze". If we can, by luck or by theory, pick a good ansatz, then we have a computable algorithm to find the solutions therein.

Organization of the Paper

Section 1 develops the ansatz. It recalls the differential polynomials in which an ansatz is written, shows how ODEs, algebraic extensions and non-polynomial functions are expressed in them, defines what we require of an ansatz, and closes with a half dozen worked examples. Section 2 presents the core algorithm, which fits a given system of differential equations to a given ansatz: a differential Thomas decomposition of the ansatz, Ritt reduction of each equation of the system modulo each component, and a projection onto the space of constants. It then proves that what the algorithm returns is exactly the set of constants for which the ansatz solves the system — neither missing valid choices nor admitting spurious ones. Section 3 illustrates the algorithm by computing a previously unknown solution to the hydrogen Schrödinger equation. Section 5 discusses how the algorithm can be generalized and section 6 discusses the theoretical issues surrounding selection of an ansatz. Section 7 outlines the software used in the example from section 3. Section 8 discusses future work and section 9 is the conclusion.

1 The Ansatz

The primary intent of the algorithm is find solutions to a PDE in the form of nested extensions, either ODE or algebraic.

We do this by introducing additional differential polynomials, parameterized with constants, forming what we term the "ansatz".

Using differential polynomials means that we can't restrict our functions to something like $L^2(\mathbb{R})$, because that requires imposing a global integral condition, and there's no way to specify that using only differential polynomials. Nor can we specify any kind of boundary condition.

We can, however, restrict the function space by requiring the solutions to satisfy additional differential equations.

1.1 Differential Polynomials

J.F. Ritt introduced differential algebra [1]. Starting with a ring R or a field F, we can construct a *differential ring* or a *differential field* by specifying one or more derivations, unary operators that obey two additional axioms:

$$dx(a+b) = dx(a) + dx(b)$$
$$dx(ab) = (dx(a))b + a(dx(b))$$

A differential ring (field) equipped with a single derivation is called an *ordinary* differential ring (field); one equipped with multiple derivations is called a *partial* differential ring (field).

Various notations are in common use for the derivations; since we work mostly in partial differential rings where there are multiple derivations, we refer to derivations by prefixing them with a d, i.e, dx. We follow Kolchin ([4] §I.1) and further assume that all derivations are pairwise commutative:

$$dx(dy(a)) = dy(dx(a))$$

Let Θ be the free abelian monoid generated by the derivations; we call the elements of Θ the *derivative operators*, and they act on the differential ring by repeated application of the derivations. The identity element of Θ acts as the identity mapping on the differential ring. The *order* of a differential operator (ord δ) is the number of derivations used to construct it.

A derivative operator applied to an indeterminate is called a *derivative*. We generally use "jet notation" for derivatives, where the applied derivations are denoted by subscripted letters on the indeterminates, i.e, Ψ_x is the result of applying derivation dx to Ψ.

It is convenient to let each indeterminate introduce an infinite set of derivatives by pairing the indeterminate with all the elements of Θ, so introducing Ψ also introduces Ψ_x, Ψ_y, Ψ_{xy}, Ψ_{xxx}, etc.

For each derivation, there will typically exist in the differential ring an associated indeterminate, labeled with the same letter, with the property that applying a derivation to its associated indeterminate produces unity, i.e, $x_x = 1$. Applying a derivation to the associated indeterminate of a different derivation produces zero, i.e, $y_x = 0$. Sometimes it is important to remember that while the same letter is used for both x and dx, they are distinct objects. x is an element of a differential ring, while dx is a unary operator defined on a differential ring.

Ritt's convention, which we follow, is to write a differential ring by stating its coefficient ring and then listing the *differential indeterminates* — the dependent variables — in curly braces: $\mathfrak{F}\{y_1, \ldots, y_n\}$ is the totality of differential polynomials in $y_1, \ldots, y_n$ over $\mathfrak{F}$ ([1], §3). The braces do *not* list the derivations, which are separate data; we write $\Delta = \{\delta_1, \ldots, \delta_m\}$ for them, and Θ for the free commutative monoid they generate, so that the elements θy_i with $\theta \in \Theta$ are the *derivatives* of the indeterminates. The independent variables belong to the coefficient ring, where they satisfy $\delta_i x_j = \delta_{ij}$ — Ritt's own convention (§29), and the one [5] and [6] follow. Thus $\mathbb{Q}[x, y, z]\{\Psi\}$ is the ring of differential polynomials in the single dependent variable Ψ, over coefficients involving the three independent variables x, y, z, with $\Delta = \{\delta_x, \delta_y, \delta_z\}$. The derivations we often write as subscripts: Ψ_x is δ_x applied to Ψ.

In the present work, primes are reserved for a different purpose — the derivatives internal to an ODE extension, where Ψ' means $d\Psi/dv$ for the extension's own independent variable v. Such a prime is never a Δ-derivative: Ψ' is a differential indeterminate in its own right, tied to Ψ by the chain rule. See the next subsection for more details.

Constants are differential indeterminates too — those that every derivation sends to zero. We name them $c_1, \ldots, c_n$ and impose $\delta_i c_j = 0$.

1.2 Describing ODEs

Ultimately, we are operating in a partial differential ring, but we are seeking to solve the PDE using ODEs. How, then, do we describe ODEs using differential polynomials? Initially, we consider homogenous linear ODEs, but this is not a fundamental limitation; see below, for example, Figure 8 and equations (25).

An n^{th}-order homogeneous linear ODE takes the form:

$$a_n(v)y^{(n)} + a_{n-1}(v)y^{(n-1)} \cdots a_1(v)y' + a_0(v)y = 0 \tag{10}$$

where $y^{(n)}, y^{(n-1)}, ...y'$ are derivatives of the dependent variable y with respect to the independent variable v. Note that all of the coefficient functions are univariate functions of v. We will further restrict them to be polynomials in v, but this is also not a fundamental limitation; see the ansätze in Figures 6 and 7 and equations (23) and (24).

To describe equation (10) using differential polynomials, we introduce $n+2$ new indeterminates. We need $n+1$ indeterminates for y and its first n derivatives, and we introduce another indeterminate for the *independent variable* v.

We create a differential polynomial in y and its derivatives whose coefficients are polynomials in v with constant coefficients (either numeric constants or constant indeterminates). Thus, the coefficients of the ODE's defining differential polynomial are selected not from the existing field, but rather from $\mathbb{Q}(v)$, an auxiliary field constructed using the independent variable v.

How do the existing derivations operate on the new indeterminates? The independent variable needs no special attention; it is the operation of the derivations on the dependent variable and its derivatives that must be specified. All that required is the chain rule. We need to specify how the new interderminates behave under the existing derivations, and that relationship is analytically:

$$\frac{\mathrm{d}\Psi}{\mathrm{d}x} = \frac{\mathrm{d}\Psi}{\mathrm{d}v}\frac{\mathrm{d}v}{\mathrm{d}x} \tag{11}$$

Algebraically, we write:

$$\Psi_x = \Psi' v_x \tag{12}$$

We use apostrophes, and not the jet notation subscripts, when writing the differential polynomial in the ODE case to emphasize that we are not introducing a new derivation, nor is the ODE's derivative any of the existing derivatives.

In short, to construct a new ODE extension:

- We introduce new indeterminates $\Psi, \Psi', ..., \Psi^{(n)}, v$
- We introduce a new differential polynomial involving only the new indeterminates and constants, and using no other non-constant indeterminates or derivatives
- We define the derivatives of $\Psi, \Psi', ..., \Psi^{(n-1)}$ using the chain rule

Example: In the differential ring $\mathbb{C}[x,y]\{\Psi,\Psi',v\}$, with $\Delta = \{\delta_x,\delta_y\}$, we can construct an ODE extension specified by the dependent indeterminate Ψ, its derivative Ψ', independent indeterminate v, and differential polynomial $\Psi' = \Psi$. This differential polynomial maps to the analytic equation $\frac{d\Psi}{dv} = \Psi$ and is solved by the analytic functions Ae^v (A is an arbitrary constant). If $v = x^2 + y$, then the functions Ae^{x^2+y} are solutions to this ODE; the differential polynomials required to model Ae^{x^2+y} are:

$$\Psi_x = 2\Psi' x \qquad \Psi_y = \Psi' \qquad \Psi' = \Psi \tag{13}$$

which, due to the equality of e^v with its own derivative, simplify to:

$$\Psi_x = 2\Psi x \qquad \Psi_y = \Psi \tag{14}$$

1.3 Describing Non-Polynomial Functions

Restricting from differential equations to differential polynomials is not a severe limitation, as so many elementary and special functions can be described as the solutions of differential polynomials. For example, both $y = \cos\theta$ and $y = \sin\theta$ are solutions to the analytic equation $y''(\theta) + y(\theta) = 0$ and can easily be introduced into a system of differential polynomials using the technique of the previous subsection.

For example, consider introducing the function $\cos x$ in the differential ring $\mathbb{Q}[x,y]\{f,f'\}$, with $\Delta = \{\delta_x,\delta_y\}$. We can replace $\cos x$ with a new indeterminate, say f, and write $f'' = -f$, where the derivative is taken with respect to x. The technique from the previous section is used to connect this with the existing derivations, as follows:

$$\begin{gathered} f_x = f' \qquad f_y = 0 \\ f'_x = f'' \qquad f'_y = 0 \\ f'' = -f \end{gathered} \tag{15}$$

This can be simplified somewhat by eliminating f'':

$$\begin{gathered} f_x = f' \qquad f_y = 0 \\ f'_x = -f \qquad f'_y = 0 \end{gathered} \tag{16}$$

The analytic solutions to these equations will be of the form $A \cos x + B \sin x$, but this does not present a serious problem. In fact, it is quite common to find a space of solutions for a differential equation, and then to impose additional restrictions (such as boundary conditions) to further restrict those solutions. Likewise, we can use f for our calculations with differential polynomials and, at the conclusion of the calculations, restrict f to be $\cos x$, not to meet a boundary condition, but to force f to be the function required by the original problem.

Likewise, $g = \exp x$ can be expressed using $g' = g$ and $h = \ln x$ using $h' = x'/x$, and the roots of polynomials are solutions to polynomial equations, so most elementary and special functions can be expressed using differential polynomials.

1.4 Forming an Ansatz

My primary goal is to "solve" a PDE, or a system of them, by finding solutions described using ODEs. Therefore, all of my current ansätze attempt to express solutions to the system using a finite combination of ODEs. They do not have to be linear, seperable or homogenous. They do require degree bounds on all of their constituent polynomials, in order to obtain a finite number of constant parameters.

Algebraic extensions can also be included, but since they must be parameterized by a finite number of constants, they can not express an arbitrary algebraic extension of indefinite degree.

To form an ansatz, we construct a tower of ODE and/or algebraic extensions that build from the base differential ring (used to write our original PDE) and culminates in a newly introduced indeterminate that is used as the equation's dependent variable. When the target is a system of interrelated equations, several parameterized differential polynomials sit at the top of the tower, each equated to a new indeterminate and used as one of the system's dependent variables (see Figure 8); the algorithm of §2 handles this case with no change beyond reducing each equation of the system in turn.

The primary requirement for an ansatz is that it be expressed using differential polynomials, as this makes it ammenable to the techniques of differential algebra. This excludes any kind of boundary conditions (which would require evaluating functions at some boundary) or Sobolov space restrictions (which would require evaluating integrals of the function), but admits almost almost any differential equation imaginable.

A good ansatz will be flexible enough to match the solution sought, but require only a small number of parameters, to make the computation tractable.

1.5 Some Sample Ansätze

There is no one correct ansatz, and currently no theory to guide their selection. Section 6 will discuss some of the difficulties developing such a theory. At the moment, ansatz selection is guided by educated guesses about what function spaces might hold interesting solutions to various PDEs.

Over the next several pages, I'll describe several possible ansätze.

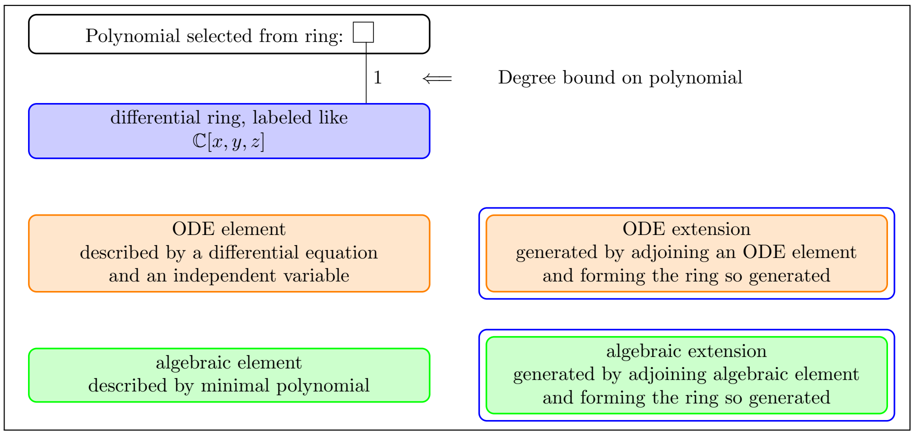

Figure 1: Key to the ansatz diagrams

We use a graphical notation to describe an ansatz, for a key see Figure 1. Blue-purple boxes denote rings, orange boxes denote ODE extensions, and green boxes denote algebraic extensions. A blue box drawn immediately around an orange or green box denote the ring formed by adjoining the element defined by the orange or green box. Polynomials are drawn as a square white box and are connected by a line to the ring to which they belong. A black number next to the line indicates a degree bound on the polynomial, and two numbers separated by a slash denote a rational function. For example, 2/2 denotes a rational function with a second degree numerator and a second degree denominator.

Two refinements of this notation are used in Figure 9, where an ansatz has been normalized and its polynomials are consequently not free. Where only some of a polynomial's coefficients remain free, the polynomial is written out in place of the single white box, with a white box standing for each coefficient that is still free: $(\square + \square s)$ is an arbitrary first degree polynomial in s, while $(1 + \square s)$ is one whose constant term has been fixed at 1. And the degree bound on a line is written only when every generator of the ring below may appear to that degree. A polynomial confined to some of the generators — an inner variable built from two of the four coordinates, say — carries no number, because no degree bound describes it; the form written out in its box is the description.

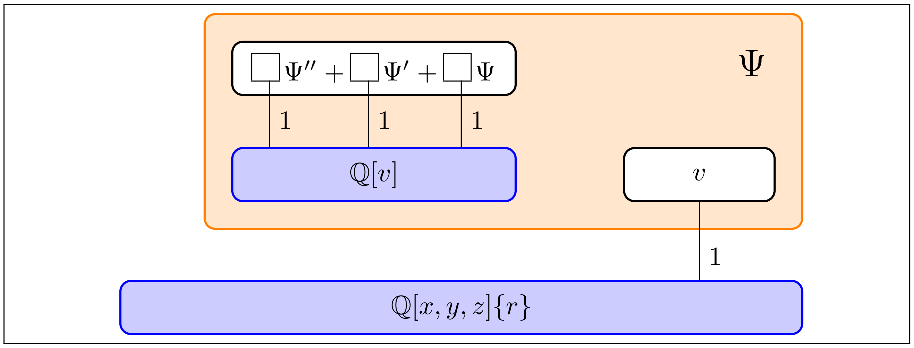

Figure 2: An ansatz for a second-order linear homogenous ODE

The ansatz diagramed in Figure 2 uses a second-order linear homogenous ODE with first degree coefficients and first degree independent variable, and is described using differential polynomials as follows.

First, a differential polynomial involving only Ψ'', Ψ', Ψ, v, and constants, and using no non-constant indeterminates or derivatives.

$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi = 0 \tag{17}$$

Next, the derivatives of Ψ and Ψ', defined using the chain rule:

$$\begin{array}{ccc} \Psi_x = \Psi' v_x & \Psi_y = \Psi' v_y & \Psi_z = \Psi' v_z \\ \Psi'_x = \Psi'' v_x & \Psi'_y = \Psi'' v_y & \Psi'_z = \Psi'' v_z \end{array} \tag{18}$$

Finally, the independent variable v:

$$v = v_1 x + v_2 y + v_3 z + v_4 r \tag{19}$$

Our differential ring is equipped with three derivations dx, dy, and dz. Our base ring is $\mathbb{Q}[x, y, z]$, and r is a differential indeterminate carrying the relation $r^2 - x^2 - y^2 - z^2$ in the system rather than a quotient taken in the ring; from this we select a linear variable v using equation (19). The v_i's (and the a_i's, b_i's and c_i's) are constants. Strictly speaking, a v_0 constant term should also be present, but we often drop the constant term from the independent variable of an ODE extension, because taking the derivative with respect to $x + 3$ is no different from taking the derivative with respect to x.

The element Ψ is expected to solve the PDE, and we use it to write the PDE.

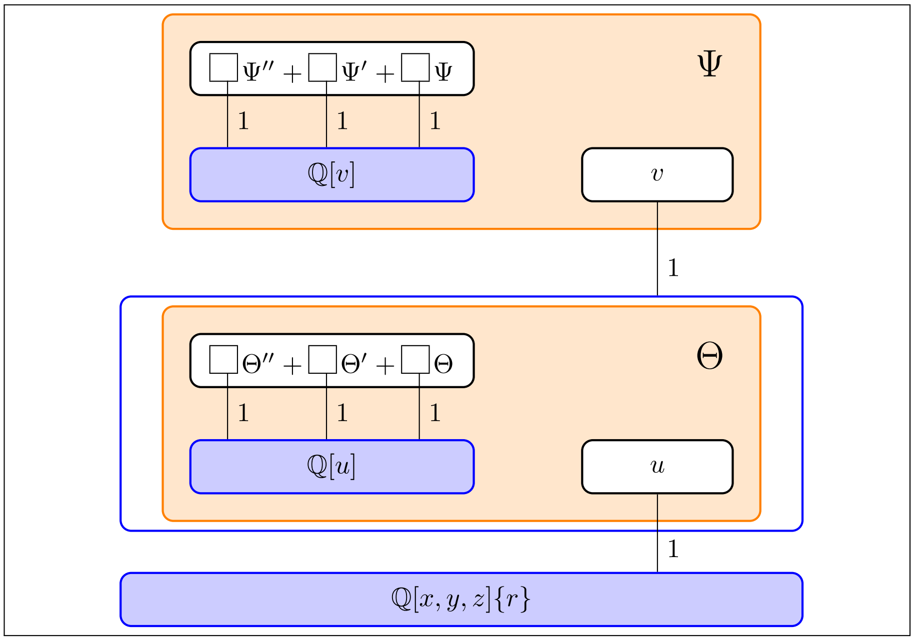

Figure 3: Two nested ODEs

Multiple extensions can be assembled together into a tower. Both ODE and algebraic extensions can be specified. Figure 3's ansatz shows two nested ODE extensions.

$$\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \tag{20a,b,c}$$

$$\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \tag{20d,e,f}$$

$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi = 0 \tag{20g}$$

$$v = v_1 x + v_2 y + v_3 z + v_4 r + v_5 \Theta \tag{20h}$$

$$\Theta_x = \Theta' v_x \qquad \Theta_y = \Theta' v_y \qquad \Theta_z = \Theta' v_z \tag{20i,j,k}$$

$$\Theta'_x = \Theta'' v_x \qquad \Theta'_y = \Theta'' v_y \qquad \Theta'_z = \Theta'' v_z \tag{20l,m,n}$$

$$(d_0 + d_1 u)\Theta'' + (e_0 + e_1 u)\Theta' + (f_0 + f_1 u)\Theta = 0 \tag{20o}$$

$$u = u_1 x + u_2 y + u_3 z + u_4 r \tag{20p}$$

$$r^2 = x^2 + y^2 + z^2 \tag{20q}$$

Again, we'll write the PDE (not shown) using the expected solution Ψ; that's the convention for the next several examples. We don't write the PDE explicitly because the ansätze can be applied to many PDEs. (25) will shown a different way to construct PDEs.

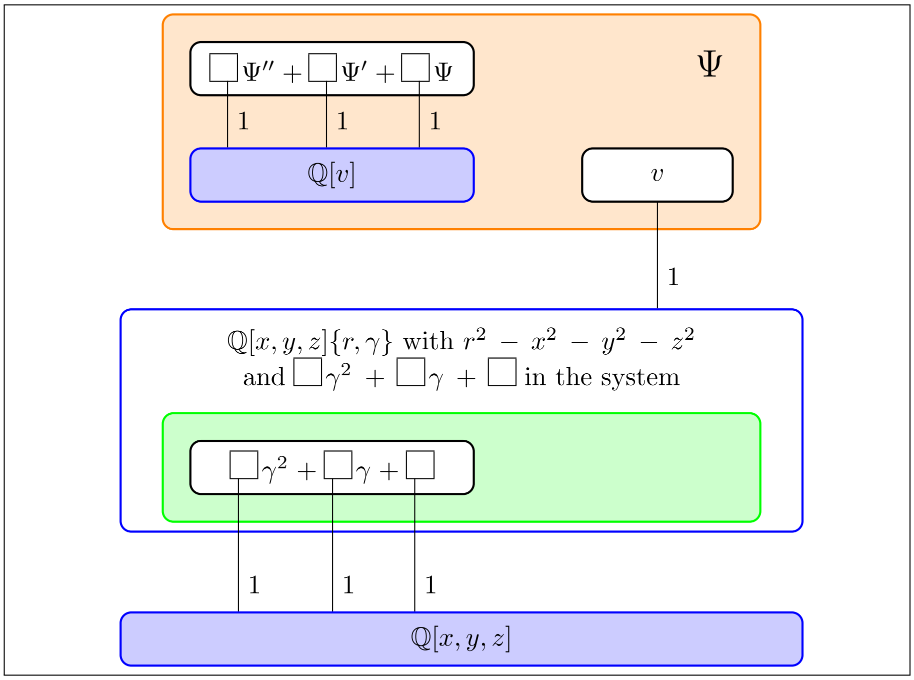

Figure 4: An ansatz for a second-order linear homogenous ODE with first degree coefficients and first degree independent variable, and an algebraic extension used in the formation of the independent variable

Multiple extensions can be assembled together into a tower. Both ODE and algebraic extensions can be specified. Figure 4's ansatz allows a second-degree algebraic extension to be used in the formation of the distinguished variable.

$$\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \tag{21a,b,c}$$

$$\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \tag{21d,e,f}$$

$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi = 0 \tag{21g}$$

$$v = (v_1 x + v_2 y + v_3 z + v_4 r + v_5 \gamma) \tag{21h}$$

$$g_0 = (g_{00} x + g_{01} y + g_{02} z + g_{03} r) \tag{21i}$$

$$g_1 = (g_{10} x + g_{11} y + g_{12} z + g_{13} r) \tag{21j}$$

$$g_2 = (g_{20} x + g_{21} y + g_{22} z + g_{23} r) \tag{21k}$$

$$g_0 \gamma^2 + g_1 \gamma + g_2 = 0 \tag{21l}$$

$$r^2 - (x^2 + y^2 + z^2) \tag{21m}$$

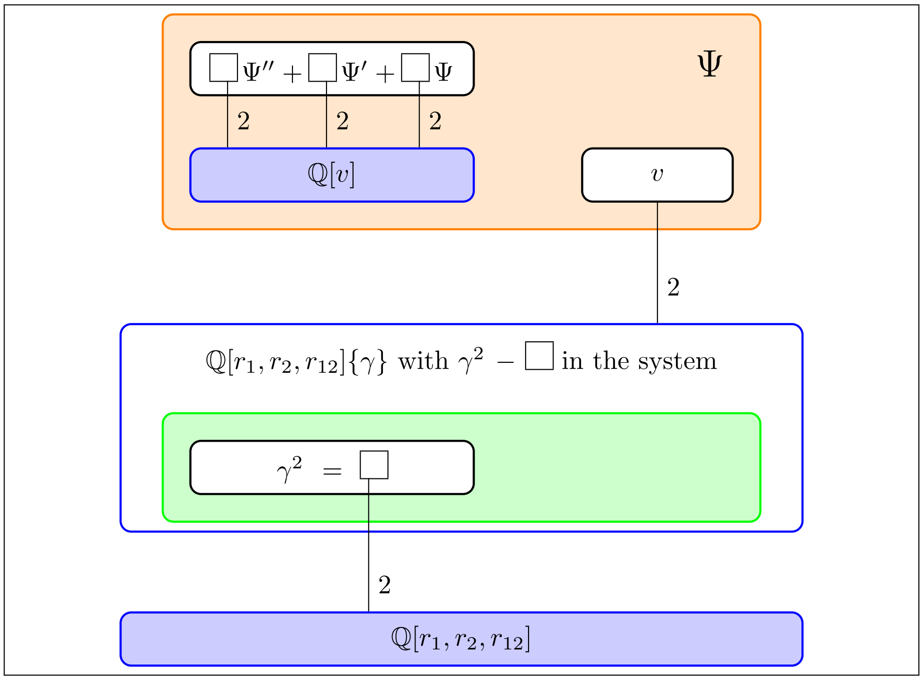

Figure 5: An ansatz for a second-order linear homogenous ODE with second degree coefficients and second degree independent variable, and a square root used in the formation of the independent variable

Second degree algebraic extensions can be specified more concisely using a square root. Higher degree parameterized polynomials can be used, second degree, in this case.

Note that a different set of derivations is used and that no algebraic extension is specified to define r, presumably because it isn't needed to construct the PDE.

$$\Psi_{r_1} = \Psi' v_{r_1} \qquad \Psi_{r_2} = \Psi' v_{r_2} \qquad \Psi_{r_{12}} = \Psi' v_{r_{12}} \tag{22a,b,c}$$

$$\Psi'_{r_1} = \Psi'' v_{r_1} \qquad \Psi'_{r_2} = \Psi'' v_{r_2} \qquad \Psi'_{r_1 2} = \Psi'' v_{r_{12}} \tag{22d,e,f}$$

$$(a_0 + a_1 v + a_2 v^2)\Psi'' + (b_0 + b_1 v + b_2 v^2)\Psi' + (c_0 + c_1 v + c_2 v^2)\Psi = 0 \tag{22g}$$

$$v = v_1 r_1 + v_2 r_2 + v_3 r_{12} + v_4 \gamma + v_5 r_1^2 + v_6 r_2^2$$

$$+ v_7 r_{12}^2 + v_8 r_1 r_2 + v_9 r_1 r_{12} + v_{10} r_2 r_{12} + v_{11} \gamma r_1 + v_{12} \gamma r_2 + v_{13} \gamma r_{12} \tag{22h}$$

$$\gamma^2 = g_0 + g_1 r_1 + g_2 r_2 + g_3 r_{12} + g_4 r_1^2 + g_5 r_2^2 + g_6 r_{12}^2$$

$$+ g_7 r_1 r_2 + g_8 r_1 r_{12} + g_9 r_2 r_{12} \tag{22i}$$

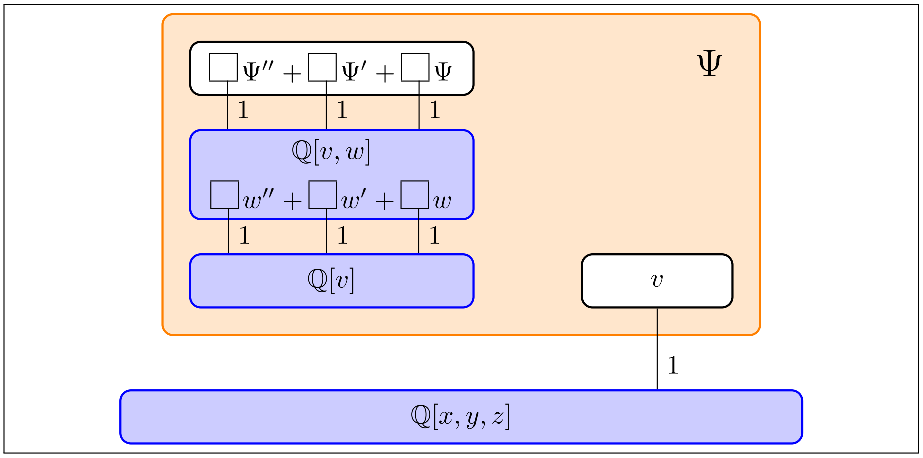

Figure 6: An ansatz with a nested ODE extension

It is also possible to introduce additional extensions into the univariate ring from which the ODE's coefficients are selected (Figure 6), i.e.:

$$
\begin{gathered}
\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \\
\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \\
(a_0 + a_1 v + a_2 w)\Psi'' + (b_0 + b_1 v + b_2 w)\Psi' + (c_0 + c_1 v + c_2 w)\Psi = 0 \\
(d_0 + d_1 v)w'' + (e_0 + e_1 v)w' + (f_0 + f_1 v)w = 0 \\
v = v_1 x + v_2 y + v_3 z
\end{gathered}
\tag{23}
$$

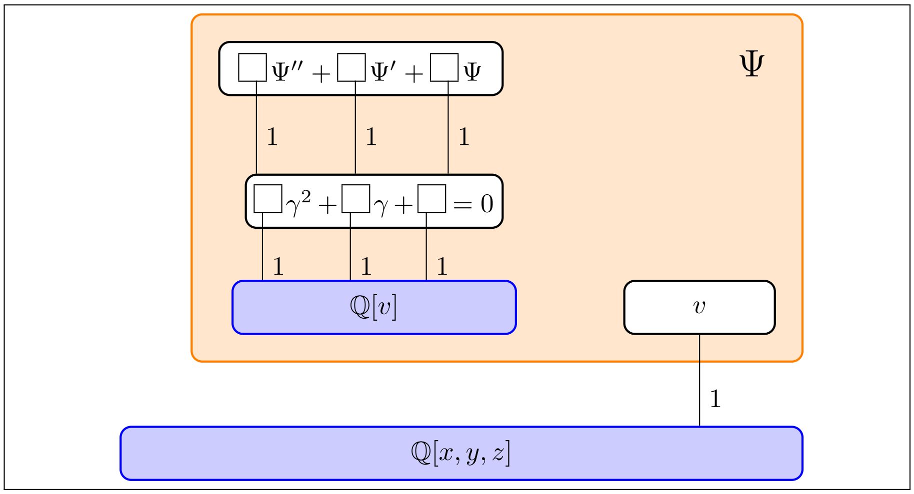

Figure 7: An ansatz with an algebraic extension in the ODE coefficient ring

Algebraic extensions can be introduced in the ODE coefficient ring.

$$
\begin{gathered}
\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \\
\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \\
(a_0 + a_1 v + a_2 \gamma)\Psi'' + (b_0 + b_1 v + b_2 \gamma)\Psi' + (c_0 + c_1 v + c_2 \gamma)\Psi = 0 \\
(d_0 + d_1 v)\gamma^2 + (e_0 + e_1 v)\gamma + (f_0 + f_1 v) = 0 \\
v = v_1 x + v_2 y + v_3 z
\end{gathered}
\tag{24}
$$

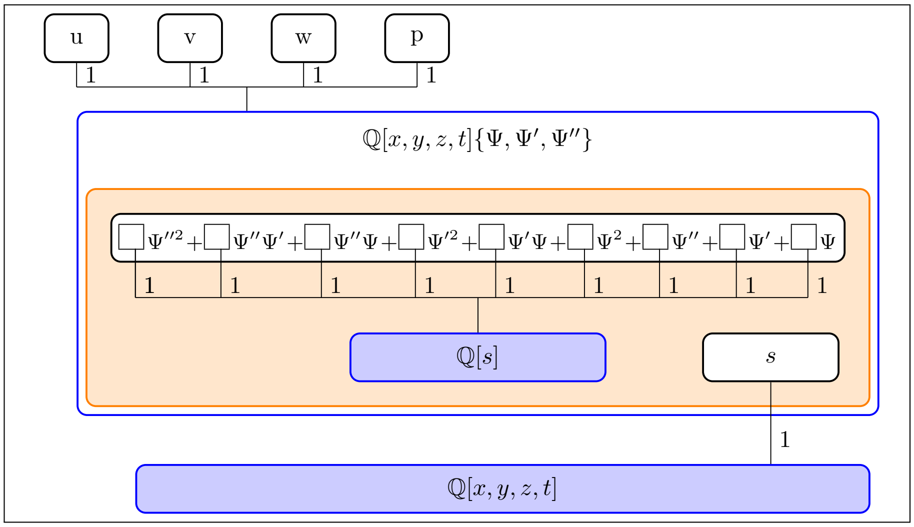

Figure 8: A nonlinear second order second degree homogeneous ansatz

Non-linear ODEs can be constructed. For a linear homogeneous second-order ODE, we have Ψ'', Ψ', and Ψ, multiplied by polynomials in the ODE's independent variable, so our differential polynomial is first degree in Ψ and its derivatives Ψ' and Ψ''. For a non-linear ODE, we can introduce higher degrees. For example, a second degree second order ODE would add Ψ''^2, $\Psi''\Psi'$, $\Psi''\Psi$, Ψ'^2, $\Psi'\Psi$, and Ψ^2 to the ansatz. See Figure 8.

This time, rather than use Ψ to solve the PDE directly, we define linear polynomials u, v, w, p that will be used to write a system of PDEs. The ODE's independent variable is called s here, rather than v as in the other ansätze, so as not to collide with the second of those four polynomials. Section 4 carries this ansatz through the incompressible Navier–Stokes equations.

$$
\begin{aligned}
&\Psi_x = \Psi' s_x \qquad \Psi_y = \Psi' s_y \qquad \Psi_z = \Psi' s_z \qquad \Psi_t = \Psi' s_t \\
&\Psi'_x = \Psi'' s_x \qquad \Psi'_y = \Psi'' s_y \qquad \Psi'_z = \Psi'' s_z \qquad \Psi'_t = \Psi'' s_t \\
&(a_0 + a_1 s)\Psi''^2 + (b_0 + b_1 s)\Psi''\Psi' + (c_0 + c_1 s)\Psi''\Psi \\
&\quad +(d_0 + d_1 s)\Psi'^2 + (e_0 + e_1 s)\Psi'\Psi + (f_0 + f_1 s)\Psi^2 \\
&\quad +(g_0 + g_1 s)\Psi'' + (h_0 + h_1 s)\Psi' + (i_0 + i_1 s)\Psi = 0 \\
&s = s_1 x + s_2 y + s_3 z + s_4 t \\
&u = u_1 x + u_2 y + u_3 z + u_4 t + u_5 \Psi \\
&v = v_1 x + v_2 y + v_3 z + v_4 t + v_5 \Psi \\
&w = w_1 x + w_2 y + w_3 z + w_4 t + w_5 \Psi \\
&p = p_1 x + p_2 y + p_3 z + p_4 t + p_5 \Psi
\end{aligned} \tag{25}
$$

2 Solution Algorithm

For a given system of differential equations, having selected an ansatz, we now seek to find the constants that allow the ansatz to solve the system, in a sense made more specific below.

2.1 Differential Algebra Preliminaries

We first recall the notions from differential algebra on which the hypotheses and theorems below rest. For the Ritt-reduction apparatus — rankings, leaders, initials and separants, reduction — our terminology and numbering follow [5], §§ 3–4. For involutivity, simple systems and the Thomas decomposition we follow [7], §§ 2–3, and for the ideal-theoretic properties of simple systems, [8], § 2.2 (see also [9], § 3).

A differential ideal is a subring of a differential ring, closed under arbitrary derivation and multiplication by arbitrary ring elements. Given a set of generators $g_1, ..., g_n$, the differential ideal they generate is denoted $[g_1, ..., g_n]$. We write $\sqrt{I}$ for the *radical* of a differential ideal I, the set of differential polynomials some power of which lies in I; in characteristic zero it is again a differential ideal.

A differential ideal I is *consistent* if $1 \notin I$; otherwise it is *inconsistent*.

Ritt defined two reduction algorithms: full and partial. We use Ritt's full reduction algorithm ([1], Chapter I, § 6; [4], Chapter I, § 9) A compact description of the relevant concepts can be found in Section 3 of [5].

Ritt's reduction algorithms require a *ranking* to be specified on the indeterminates and their derivatives. This is a distinct concept from an ordering on the monomials, as used, for example, in defining a Gröbner basis, and is more in the spirit of triangular decomposition algorithms than Gröbner basis algorithms.

A ranking on a differential ring R is an ordering on the ring's indeterminates and their derivatives that meets two compatibility conditions:

$$\begin{aligned} \forall \delta \in \Theta, \forall u, v \in R \quad u < v &\implies \delta u < \delta v \\ \forall \delta \in \Theta, \forall u \in R \quad u &< \delta u \end{aligned} \tag{26}$$

The standard definition orders only the derivatives of the differential indeterminates. We follow [10] in extending a ranking to the independent variables of the coefficient ring as well, placing them in the lowest block; nothing in the theory depends on this, but it lets a single ordering be written for every indeterminate the computation touches.

A ranking is said to be *orderly* if meets the additional condition:

$$\forall \delta, \theta \in \Theta, \forall u, v \in R \quad \text{ord } \delta < \text{ord } \theta \implies \delta u < \theta v \tag{27}$$

Note that an orderly ranking ranks all pure indeterminates lower than any proper derivative.

The *leader* of a polynomial is the highest ranking indeterminate or derivative that appears in the polynomial with a non-zero coefficient. Writing u_p for the leader of p, the *initial* of p is its leading coefficient regarded as a univariate polynomial in u_p, and the *separant* of p is $s_p = \partial p / \partial u_p$. For a set $A \subseteq R \setminus K$ we write I_A, S_A, and $H_A = I_A \cup S_A$ for the sets of initials, separants, and both.

A differential polynomial p is said to be *partially reduced* with respect to a differential polynomial q if no proper derivative of q's leader appears in p, and *fully reduced* if it is partially reduced and q's leader, if it appears in p, appears at a lower degree than its degree in q.

Ritt's full reduction algorithm reduces a differential polynomial p by a system S of differential polynomials and produces differential polynomials r (the remainder) and $h \in H_S$ such that r is fully reduced with respect to all of the differential polynomials in S and $(hp - r)$ is an element of the differential ideal $[S]$ generated by the elements of S and all their derivatives. Put another way, r is an element of the saturated differential ideal $[S] : H_S^\infty$.

A set $A \subseteq R \setminus K$ is *triangular* if the leaders of its elements are pairwise distinct, and *differentially triangular* ([5], Definition 13) if in addition its elements are pairwise partially reduced.

The differential Thomas decomposition rests on a constructive completion condition, phrased through an involutive division. Fix an ordering $d_1 < \cdots < d_n$ of the derivations. Given a finite set U of derivatives, write each $u \in U$ as θv for an indeterminate v and $\theta = d_1^{\mu_1} \cdots d_n^{\mu_n} \in \Theta$, and call $\mu(u) = (\mu_1, \ldots, \mu_n)$ its multi-index. *Janet's division* declares d_j *multiplicative* for $u \in U$ if $\mu_j(u)$ is maximal among the multi-indices of those elements of U that share the indeterminate v and agree with u in $\mu_1, \ldots, \mu_{j-1}$; the remaining derivations are *non-multiplicative* for u. A *non-multiplicative prolongation* of a differential polynomial p is $d_j\, p$ for a derivation d_j that is non-multiplicative for the leader of p.

A differential system $S = (S^=, S^{\neq})$ of equations $S^=$ and inequations $S^{\neq}$ is *(Janet-)involutive* ([7], Definition 3.5) if every non-multiplicative prolongation of an element of $S^=$ reduces to zero modulo $S^=$.

Alongside this differential condition sit three algebraic ones. A system S, regarded as a system in the finitely many differential variables that occur in it, is *algebraically simple* ([7], Definition 2.2) if

(A1) S is *triangular*: no two elements of S share a leader, and no element of S lies in the base field;

(A2) S has *non-vanishing initials*: for each equation p of S, with leader x, the initial of p is non-zero at every solution of the subsystem of S in the variables ranked strictly below x; and

(A3) S is *square-free*: for each equation p of S, with leader x, the univariate polynomial in x obtained by evaluating the coefficients of p at a solution of that lower subsystem is square-free, for every such solution.

Conditions (A2) and (A3) are quantified over the solutions of the lower part of the system rather than imposed at a single point. That is what makes them stable under specialization of lower-ranked variables, and it is the fact on which Lemma 4 turns. Condition (A3) also has a consequence we use without further comment: a square-free polynomial and its separant have no common root, so the separants of S, like its initials, do not vanish on the solutions of the lower-ranking subsystems ([7], § 2, footnote 3).

A system $S = (S^=, S^{\neq})$ is *differentially simple* ([7], Definition 3.5) if

(S1) S is algebraically simple in the finitely many differential variables that appear in it;

(S2) S is involutive;

(S3) $S^=$ is minimal; and

(S4) no inequation of $S^{\neq}$ is reducible modulo $S^=$.

A decomposition of a system into finitely many differentially simple subsystems whose solution sets are pairwise *disjoint* is a *differential Thomas decomposition.* [7] gives an algorithm computing one, and [8] develops the theory at length; [9] is a shorter and freely available account.

Everything the completeness argument of §2.4 requires is supplied by a single result about simple systems, which we state in the form we use. Write q for the product of the initials and separants of the equations of $S^=$.

Proposition 1 ([8], Prop. 2.2.50; [9], Prop. 3.31)**.** *Let S be a simple differential system with equations $p_1 = 0, \dots, p_s = 0$, let $E = [p_1, \dots, p_s]$ be the differential ideal they generate, and let q be the product of the initials and separants of $p_1, \dots, p_s$. Then $E : q^\infty$ is a radical differential ideal, and for $p \in R$,*

$$p \in E : q^\infty \iff \text{the pseudo-remainder of } p \text{ modulo } p_1, \dots, p_s \text{ and their derivatives is zero.}$$

Two features of this statement matter here. It is an *equivalence*, so reduction to zero is a decision procedure and not merely a sufficient condition; and its hypothesis is simplicity alone, so no separate coherence, regularity or characteristic-set hypothesis is needed. The same proposition is restated, with proof references, as Proposition 2.29 of [11].

The companion statement assembles the components of a decomposition. For a system S that is *not* assumed simple, with Thomas decomposition $S_1, \dots, S_s$, writing $E^{(i)}$ for the differential ideal generated by the equations of S_i and $q^{(i)}$ for the corresponding product of initials and separants ([8], Prop. 2.2.72; [9], Prop. 3.32),

$$\sqrt{E : q^\infty} \;=\; \big(E^{(1)} : (q^{(1)})^\infty\big) \cap \cdots \cap \big(E^{(s)} : (q^{(s)})^\infty\big).$$

A differential polynomial is therefore a consequence of the whole system exactly when it reduces to zero modulo every component of the decomposition.

2.2 The Constant Loci

We wish to solve a system of differential equations, where the solution must be of a restricted form, parameterized by constants, the *ansatz.*

Given an ansatz $A(c^*) = \{a_1(c^*), ..., a_k(c^*)\}$, a collection of differential polynomials parameterized by constants $c^* \in \mathbb{C}^n$, and a second collection $\mathcal{P} = \{P_1, \ldots, P_t\}$ — the system we wish to solve — we now wish to find values for the constants that will solve the system.

We write $\mathcal{P}$ for the target system, reserving S for a system in the sense of §2.1, which carries inequations as well as equations. Nothing in what follows requires the elements of $\mathcal{P}$ to be partial differential equations. An ordinary differential equation, or a purely algebraic relation among the dependent variables, is a differential polynomial like any other, and the development never asks which kind it is; all that is used is that $\mathcal{P}$ is a finite subset of R. The case $t = 1$ recovers the single-PDE problem treated in §3; the Navier–Stokes example of §4 has $t = 4$.

It is convenient to define three constant loci:

$$V_{\exists} = \{c^* \in \mathbb{C}^n : 1 \notin [A(c^*), \mathcal{P}]\} \tag{28}$$

The *existence locus* is the set of constants that make the differential ideal $[A(c^*), \mathcal{P}]$ — generated by the ansatz together with every element of the system — consistent.

$$V_{\exists\setminus\Psi} = \{c^* \in \mathbb{C}^n : 1 \notin [A(c^*), \mathcal{P}] : \Psi^\infty\} \tag{29}$$

A differential polynomial p is *homogenous* w.r.t. an element Ψ if every term of p contains exactly one derivative (not necessarily proper) of Ψ to first degree. If every element of $\mathcal{P}$ is homogenous w.r.t. Ψ, then we define the *homogenous existence locus* as the set of constants that make the differential ideal $[A(c^*), \mathcal{P}] : \Psi^\infty$ consistent.

$$V_{\forall} = \{c^* \in \mathbb{C}^n : \mathcal{P} \subseteq \sqrt{[A(c^*)]}\} \tag{30}$$

The *membership locus* is the set of constants for which every element of $\mathcal{P}$ is in the radical differential ideal defined by the ansatz, meaning that any function satisfying the ansatz will satisfy every equation of the system. The radical is the right ideal to take here: it is the vanishing ideal of the ansatz's solution set, so membership in it says exactly what the words "any function satisfying the ansatz" say, whereas membership in $[A(c^*)]$ itself is a strictly stronger and less natural condition.

The membership locus is the one place where the passage from a single equation to a system is entirely painless. Membership in an ideal is a condition on one differential polynomial at a time, so requiring it of every element of $\mathcal{P}$ is a conjunction of t conditions of exactly the form already studied, and

$$V_\forall \;=\; \bigcap_{j=1}^{t} \{\, c^* \in \mathbb{C}^n : P_j \in \sqrt{[A(c^*)]}\,\}. \tag{31}$$

Everything below — the reduction, the projection, the completeness theorem — is applied to each P_j separately, and the results are intersected. The existence loci are not decomposable in this way, since consistency of $[A(c^*), \mathcal{P}]$ is a property of the ideal the whole system generates and is not implied by the consistency of its parts; that asymmetry is another reason to prefer the membership locus.

Often we are interested in finding the existence locus, but if every element of $\mathcal{P}$ is homogenous (and the ansatz too), then $V_\exists$ is all of $\mathbb{C}^n$, since the zero function will solve a homogenous differential equation for any value of the constants, in which case the homogenous existence locus is the more natural objective.

However, computing $V_\exists$ or $V_{\exists\setminus\Psi}$ can present serious computational challenges, so for practical reasons it is often useful to work with the membership locus, which keeps the system out of the ideal's generators.

The three constant loci are interrelated as follows:

$$V_\forall \subseteq V_{\exists\setminus\Psi} \subseteq V_\exists \tag{32}$$

Proof: an ideal is always contained in its saturation, so, for a given set of constants c^*,

$$\begin{gathered} 1 \in [A(c^*), \mathcal{P}] \implies 1 \in [A(c^*), \mathcal{P}] : \Psi^\infty \\ 1 \notin [A(c^*), \mathcal{P}] : \Psi^\infty \implies 1 \notin [A(c^*), \mathcal{P}] \\ V_{\exists\setminus\Psi} \subseteq V_\exists \end{gathered} \tag{33}$$

Note: We introduce the assumption here that the ansatz (by itself) is consistent for all values of c^*.

Furthermore, if every element of $\mathcal{P}$ lies in $\sqrt{[A(c^*)]}$ then so does the whole ideal $[A(c^*), \mathcal{P}]$, since $\sqrt{[A(c^*)]}$ is a differential ideal containing both $A(c^*)$ and $\mathcal{P}$. Hence

$$\begin{gathered} \mathcal{P} \subseteq \sqrt{[A(c^*)]} \implies [A(c^*), \mathcal{P}] \subseteq \sqrt{[A(c^*)]} \\ \mathcal{P} \subseteq \sqrt{[A(c^*)]} \wedge 1 \notin [A(c^*)] \implies 1 \notin [A(c^*), \mathcal{P}] \\ V_\forall \subseteq V_\exists \end{gathered} \tag{34}$$

the second line because $1 \in \sqrt{I}$ if and only if $1 \in I$.

If the ansatz does not force Ψ to vanish — that is, if $\Psi \notin \sqrt{[A(c^*)]}$ — then saturating the containment above by Ψ^∞ gives

$$\begin{gathered}
\mathcal{P} \subseteq \sqrt{[A(c^*)]} \implies [A(c^*), \mathcal{P}] : \Psi^\infty \subseteq \sqrt{[A(c^*)]} : \Psi^\infty \\
\Psi \notin \sqrt{[A(c^*)]} \implies 1 \notin \sqrt{[A(c^*)]} : \Psi^\infty \\
V_\forall \subseteq V_{\exists \setminus \Psi}
\end{gathered} \tag{35}$$

the middle line because $1 \in \sqrt{[A(c^*)]} : \Psi^\infty$ means $\Psi^k \in \sqrt{[A(c^*)]}$ for some k, and a radical ideal containing Ψ^k contains Ψ. This is the one place where the radical formulation pays a visible dividend: the non-degeneracy condition, which for the non-radical ideal had to be written $1 \notin [A(c^*)] : \Psi^\infty$, becomes the transparent statement that the ansatz does not force $\Psi \equiv 0$.

These relationships are illustrated by the following example.

Example 2. In the differential ring $\mathbb{C}[t]\{\Psi, c\}$, with the single derivation δ_t and $\delta_t c = 0$, consider the ansatz

$$A(c) = \{\Psi_{tt} - c\Psi\} \tag{36}$$

and the one-element system $\mathcal{P} = \{P\}$ with

$$P = \Psi_t - \Psi \tag{37}$$

To compute $V_\exists$, we need to identify the constants where $1 \notin [A(c), P]$. For any constant c, the leaders of $\{A(c), P\}$ are Ψ_t (order 1) and Ψ_{tt} (order 2). Elements in a differential ideal are constructed by multiplying by polynomials or applying derivation operators, neither of which reduce the order of a polynomial (does the ranking have to be orderly for this to be true?). Therefore, no polynomial of order zero can appear in $\{A(c), P\}$ and, in particular, $1 \notin [A(c), P]$, so $V_\exists = \mathbb{C}$.

To compute $V_{\exists \setminus \Psi}$, we need to identify the constants where $1 \notin [A(c), P] : \Psi^\infty$.

To compute $V_\forall$, we need to identify the constants where $P \in \sqrt{[A(c)]}$. Since $A(c)$ has order 2 for all values of c, every element of $[A(c)]$ has order at least 2, while every power of P has order 1. So no power of P lies in $[A(c)]$, hence $P \notin \sqrt{[A(c)]}$ for any value of c, and $V_\forall = \emptyset$.

$$V_\exists = \mathbb{C} \qquad V_{\exists \setminus \Psi} = \{1\} \qquad V_\forall = \emptyset$$

□

It is perhaps easiest to understand this example analytically. For any constant c, we can set $\Psi = 0$ and get a solution to both the ansatz and the PDE, so $V_\exists = \mathbb{C}$. If $c = 1$, then ae^t is a solution to both $A(1)$ and P, and 1 is the only value of c for which they have a common non-zero solution, so $V_{\exists\setminus\Psi} = \{1\}$. However, although both ae^t and ae^{-t} solve $A(1)$, only ae^t solves P, so a solution to the ansatz does not necessarily solve the PDE and $V_\forall = \emptyset$.

The most straightforward way to compute the existence loci $V_\exists$ or $V_{\exists\setminus\Psi}$ is to proceed as their definitions suggest: combine the system and the ansatz together as the generators of a singular differential ideal. We do not wish to treat the constants as indeterminates (as they would always be non-zero), but rather as parameters in the base field, and seek conditions on those parameters that would yield a non-trivial differential ideal where $1 \notin [A(c^*), \mathcal{P}]$ or $1 \notin [A(c^*), \mathcal{P}] : \Psi^\infty$. This suggests the use of a parametric algorithm, such as parametric Rosenfeld-Gröbner [6] or a differential Thomas decomposition [7, 8]. The author's practical experience, however, is that this approach tends to be computationally infeasible.

Therefore, for the remainder of this paper, we will focus our attention on the membership locus $V_\forall$. This restriction allows us to develop an algorithm that, while still using a parametric decomposition, only does so on a differential ideal formed from the ansatz, with the system treated separately. It is also what makes the size of the system almost costless: the expensive step, the decomposition, is performed on the ansatz alone and is therefore independent of t, and each additional equation adds only a Ritt reduction and its coefficients. This tends to be far more computationally feasible than trying to compute the existence loci $V_\exists$ or $V_{\exists\setminus\Psi}$.

2.3 Computing the Membership Locus

Given an ansatz $A(c^*)$, we first compute a differential Thomas decomposition of the ansatz, which produces a partition of the constant space into quasi-projective varieties (i.e, defined by both equations and inequations), each equiped with a specialization of the ansatz in a differentially simple form. For each differentially simple subsystem in the decomposition, we then proceed as follows.

We use Ritt's full reduction algorithm to reduce each element of the system modulo the subsystem, obtaining remainders $r_1, \ldots, r_t$, and project them onto the space of the n constant parameters $c = (c_1, \ldots, c_n)$.

To perform the projection, it is useful to classify the indeterminates in our differential ring as follows: - constants - dependent variables and their jet coordinates - independent variables - algebraic functions of the independent variables

Membership of P_j in $\sqrt{[A(c^*)]}$ is a statement about r_j as a polynomial, not about its value at any particular point: it requires r_j to vanish identically once the constants are fixed. We therefore treat all of the non-constant indeterminates alike, making no distinction between the independent variables, the dependent variables and their jet coordinates.

Accordingly, we split each term of each reduced equation r_j into a power product in the non-constant indeterminates and a coefficient in the constants alone, gathering the terms that share a power product:

$$r_j \;=\; \sum_{\alpha=1}^{N_j} m_{j\alpha}\, p_{j\alpha}(c), \qquad j = 1, \ldots, t, \tag{38}$$

where the $m_{j\alpha}$ are the N_j distinct power products occurring in r_j and each coefficient $p_{j\alpha}(c)$ is a polynomial in the constants.

The expansion (38), and its uniqueness, amount to no more than the observation that R is a polynomial ring in the non-constant indeterminates over $\mathbb{Q}[c_1, \ldots, c_n]$: the constants are sent to zero by every derivation, so they contribute no derivatives, while the x_j, the u_l and every derivative of the u_l are algebraically independent over them. It will be convenient to name the set of coefficients the expansion produces, and to carry in the notation the ring they are taken from.[1] Let $A \subseteq R$ be a subring over which R is a polynomial ring in the remaining indeterminates, so that each $r \in R$ has a unique expansion $r = \sum_{\alpha=1}^{N} m_\alpha\, p_\alpha$ with the m_α distinct power products in those indeterminates and $p_\alpha \in A$; put

$$\operatorname{Coeffs}(r, A) \;=\; \{\, p_1, \ldots, p_N \,\} \;\subset\; A. \tag{39}$$

The only case we need is $A = \mathbb{Q}[c_1, \ldots, c_n]$, for which the remaining indeterminates are exactly the non-constant ones and $\operatorname{Coeffs}(r_j, \mathbb{Q}[c_1, \ldots, c_n])$ is the coefficient set $\{p_{j1}, \ldots, p_{jN_j}\}$ of (38). Naming the ring is not pedantry here: the whole force of the projection step is that the coefficients are taken over the constants and nothing else, so that the resulting conditions are conditions on c alone. The coarse variant discussed below is, in this notation, the one that takes $A = \mathbb{Q}[c_1, \ldots, c_n]\{u_1, \ldots, u_k\}$ instead.

Note that Coeffs is a syntactic operation on a polynomial rather than a computation — reading off coefficients from a presentation the ring already has —

[1] The operation is standard even though the notation for it is not. It is Boulier's `Coeffs` ([10], `DifferentialAlgebra[Tools][Coeffs]`), whose optional `basefield` argument splits an equation so that "the `C[i]` only depend on variables which belong to the field, and the `M[i]` only depend on variables which do not" — our second argument exactly. In the cylindrical-algebraic-decomposition literature it is written coeff, the split being named there by a main variable rather than by a subring, and extended to a family of polynomials by the same union we take in (40): compare McCallum's projection operator $P(A) = \operatorname{cont}(A) \cup \operatorname{coeff}(B) \cup \operatorname{disc}(B) \cup \operatorname{res}(B)$ ([12], § 2.1). The trailing s distinguishes the set of all coefficients from the coefficient of a single monomial, as it does in Maple's `coeff` versus `coeffs`.

and that $\mathrm{Coeffs}(0, A) = \emptyset$, the case of an equation that reduces to zero on the component and so imposes no condition there.

Distinct monomials in a polynomial ring are linearly independent over $\mathbb{C}$, so r_j vanishes identically at c^* if and only if every one of its coefficients does. The projection step therefore replaces the reduced system $r_1, \ldots, r_t$ by the single system

$$\begin{gathered} p(c) = 0 \quad \text{for every } p \in J, \\ J \;=\; \bigcup_{j=1}^{t} \mathrm{Coeffs}\big(r_j,\ \mathbb{Q}[c_1, \ldots, c_n]\big) \end{gathered} \tag{40}$$

of $N_1 + \cdots + N_t$ polynomial equations in the constants alone. No elimination is required: because the coefficients already involve nothing but the constants, the system is a description of the locus in $\mathbb{C}^n$ directly.

This pooling of the coefficients is the whole of what a system demands. The equations of $\mathcal{P}$ must hold simultaneously, so the conditions they impose on the constants are conjoined; and since each r_j is required to vanish identically, the conjunction is nothing more than the union of the coefficient sets. The power products are collected within each remainder and not across remainders: two different r_j may well share a power product, but their coefficients are separate conditions and must not be added together. Nothing else in the treatment is sensitive to t.

An earlier form of this algorithm split the terms more coarsely — a power product in the independent variables and their algebraic functions, against a coefficient in the constants, the dependent variables and their jets — and then projected onto the constants with an elimination ideal. That variant is cheaper, since it produces fewer and lower-degree coefficients, and it remains *complete*: a c^* at which r_j vanishes identically certainly admits some assignment of the jet coordinates killing the coefficients. But the converse fails, because an assignment may exist without $r_j(c^*)$ being the zero polynomial, so the elimination variant returns a superset of the membership locus. The splitting above is the one for which Theorem 5 is an equivalence. A second reason to prefer it appears only for a system: under the coarse splitting the jet coordinates are existentially quantified, and quantifying them once per equation is weaker than quantifying them once for the system, so the elimination variant would have to choose between two inequivalent readings. Requiring each r_j to be the zero polynomial makes the question moot.

Finally, the remaining system defines an algebraic variety

$$\mathcal{V} \;=\; V(J) \;\subseteq\; \mathbb{C}^n,$$

which need not be irreducible. Because only the zero locus matters, the decomposition we want is into irreducible components, and these are given by the minimal associated primes: under the well-known correspondence between prime ideals and irreducible varieties ([13], Corollary 4.5.4), the minimal primes of the ideal generated by J cut out the irreducible components of $\mathcal{V}$, and several algorithms have been proposed and implemented to compute them ([2], among others). The radical never has to be formed: an ideal and its radical have the same minimal primes, so the decomposition may be run on the ideal as it stands.

To summarize, we record the algorithm. Since an ideal is an infinite object, every ideal it manipulates is carried as a finite generating set: we write $\langle F\rangle$ for the ideal generated by a finite set $F \subset \mathbb{Q}[c_1,\ldots,c_n]$ and $V(F)$ for its zero set in $\mathbb{C}^n$. A pair (F,h) consisting of such a set and a polynomial $h \in \mathbb{Q}[c_1,\ldots,c_n]$ denotes the constructible set $V(F)\setminus V(h) \subseteq \mathbb{C}^n$, and a finite set of such pairs denotes the union of the sets its members denote. $\mathrm{Coeffs}(r,A)$ is the coefficient set of (39), its second argument naming the ring the coefficients are taken over; it is $\mathbb{Q}[c_1,\ldots,c_n]$ everywhere below, and displaying it is what makes lines 6, 8 and 9 visibly the three places where the computation descends to the constants. $S_i^{=}$ and $S_i^{\neq}$ are the equations and inequations of the i^{th} component, treated as sets of differential polynomials so that they may be intersected with a subring.

Nothing in the algorithm is new. It calls four subroutines, each taken from the literature and used unmodified; the work is in what is handed to them and how their results are combined. We name them as their sources do, and give the source's own numbering where the source numbers its algorithms. Coeffs, $\cap$ and $\prod$ are not among them: those steps are syntactic, and it is worth keeping them visibly distinct from the four that do real work.

subroutine	what it computes	source
DifferentialDecompose	a differential Thomas decomposition: a partition of the solution set of a differential system into finitely many differentially simple subsystems	[7], Alg. 3.6
FullReduce	Ritt's full reduction: the remainder r, fully reduced with respect to the equations of the system. Because the system is simple, r is zero precisely for the members of the saturated ideal (Proposition 1)	[1], §6
minAss	generating sets for the minimal associated primes of an ideal presented by generators	[2], Cor. 3.2(v), §9
SMPD	*strongly make pairwise disjoint*: replaces a family of pieces by pairwise disjoint pieces, each member of the family being a union of some of them	[14], Alg. 3

Algorithm 3 (MembershipLocus)**.**

Input: A differential polynomial ring $R = \mathbb{Q}[x_1, \ldots, x_m]\{u_1, \ldots, u_k, c_1, \ldots, c_n\}$ with derivations $\Delta = \{\delta_1, \ldots, \delta_m\}$ satisfying $\delta_i x_j = \delta_{ij}$ and $\delta_i c_l = 0$; a finite system $\mathcal{P} = \{P_1, \ldots, P_t\} \subset R$ of differential polynomials; an ansatz $A(c) \subset R$, a finite set of differential polynomials parameterized by the constants $c = (c_1, \ldots, c_n)$; a block ranking $\prec$ with

$$\{u_1, \ldots, u_k\} \gg \{c_1, \ldots, c_n\} \gg \{x_1, \ldots, x_m\};$$

and a flag $\mu \in \{\forall, \exists\forall\}$ selecting the assembly.

Output: A constructible subset of $\mathbb{C}^n$, presented as a finite set of pairs $(\mathfrak{p}, h)$ in which $\mathfrak{p}$ generates a prime ideal, and so as a finite union of irreducible locally closed sets. For $\mu = \forall$ it is the membership locus $V_\forall = \{\, c^* \in \mathbb{C}^n : \mathcal{P} \subseteq \sqrt{[A(c^*)]} \,\}$. For $\mu = \exists\forall$ it is the intermediate locus $V_{\exists\forall}$, which satisfies $V_\forall \subseteq V_{\exists\forall} \subseteq V_\exists$ whenever the ansatz is consistent throughout $\mathbb{C}^n$ — a condition the algorithm reports, as described below.

Algorithm:

1: $(S_1, \ldots, S_s) \leftarrow \mathrm{DifferentialDecompose}(A(c), \prec)$
2: **for** $i = 1, \ldots, s$ **do**
3: $\quad J_i \leftarrow \emptyset$
4: $\quad$ **for** $j = 1, \ldots, t$ **do**
5: $\qquad r \leftarrow \mathrm{FullReduce}(P_j, S_i^{=})$
6: $\qquad J_i \leftarrow J_i \cup \mathrm{Coeffs}\big(r,\ \mathbb{Q}[c_1, \ldots, c_n]\big)$
7: $\quad$ **end for**
8: $\quad E_i \leftarrow S_i^{=} \cap \mathbb{Q}[c_1, \ldots, c_n]$
9: $\quad h_i \leftarrow \prod\big(S_i^{\neq} \cap \mathbb{Q}[c_1, \ldots, c_n]\big)$
10: $\quad C_i \leftarrow (E_i, h_i)$
11: $\quad W_i \leftarrow \big\{\, (\mathfrak{p}, h_i) \ :\ \mathfrak{p} \in \mathrm{minAss}\big(\langle J_i \cup E_i \rangle : h_i^\infty\big) \,\big\}$
12: **end for**
13: **if** $\mu = \forall$ **then**
14: $\quad \mathcal{B} \leftarrow \mathrm{SMPD}(\mathbb{C}^n, W_1, C_1, \ldots, W_s, C_s)$
15: $\quad D \leftarrow \bigcup\big\{\, B \in \mathcal{B} \ :\ B \subseteq W_i \text{ for every } i \text{ with } B \subseteq C_i \,\big\}$
16: **else**
17: $\quad D \leftarrow \bigcup \mathrm{SMPD}(W_1, \ldots, W_s)$
18: **end if**
19: **return** $\big\{\, (\mathfrak{p}, h) \ :\ (F, h) \in D,\ \mathfrak{p} \in \mathrm{minAss}(\langle F \rangle : h^\infty) \,\big\}$

Line 5 is the differential elimination step and line 6 the $\forall$ projection step; the inner loop of lines 4–7 performs them once for each equation of the system and accumulates the results in a single J_i, which is the only respect in which a system differs from a single equation. Lines 8 and 9 are set intersections with a subring, no computation being needed to recognize a polynomial in the constants alone; the empty product at line 9 is 1, which is the case of a component carrying no constant inequation. That these intersections are exactly the component's condition on the constants is the content of the discussion following Corollary 6. Writing $\mathcal{V}_i := V(J_i)$ for the variety of the projection and $C_i = V(E_i) \setminus V(h_i)$ for the *cell* of the component, we have $W_i = \mathcal{V}_i \cap C_i$, and the assemblies of lines 15 and 17 are

$$V_\forall \;=\; \bigcap_{i=1}^{s} \big(W_i \cup (\mathbb{C}^n \setminus C_i)\big), \qquad V_{\exists\forall} \;=\; \bigcup_{i=1}^{s} W_i :$$

under $\mu = \forall$ a point qualifies exactly when it lies in $\mathcal{V}_i$ for *every* component whose cell contains it, and under $\mu = \exists\forall$ as soon as it lies in $\mathcal{V}_i$ for *some* such component.

Lines 1–12 are common to both assemblies, as is the normalization in line 19; only lines 13–18 differ. The reason is that $\mathcal{V}_i \cup (\mathbb{C}^n \setminus C_i) = W_i \cup (\mathbb{C}^n \setminus C_i)$: outside its own cell a component imposes no condition at all, so each $\mathcal{V}_i$ is consulted only where its cell lies, and $W_i = \mathcal{V}_i \cap C_i$ is the only thing either assembly reads. The decomposition into irreducible pieces therefore belongs inside the loop, where it is done once per component, rather than after the assembly.

Two remarks on line 11. Saturating by h_i^∞ deletes the components of $V(J_i \cup E_i)$ that lie wholly inside the inequation locus, but it does not discharge the inequations: $V(\mathfrak{a} : h^\infty)$ is the Zariski *closure* of $V(\mathfrak{a}) \setminus V(h)$, so each surviving prime still carries $h_i \neq 0$ as a residual condition cutting a proper closed subset out of it. Each piece is accordingly carried as a pair — a prime and an inequation — and it is this, not anything in the assembly, that makes the output constructible rather than a variety. Second, the radical of $\langle J_i \cup E_i \rangle$ is never formed, for the reason given above: an ideal and its radical have the same minimal primes.

The $\forall$ assembly is an intersection, not a union. A component contributes a condition only where its cell lies, but where it does lie the condition must be met: if c^* falls in the cells of two components and the ansatz fails to solve the system on the second, then $A(c^*)$ has solutions that do not satisfy every element of $\mathcal{P}$, and c^* is not in the membership locus. Since the cells need not be disjoint (Corollary 6), the result is in general constructible rather than a variety. Only lines 1 and 5 involve differential algebra; lines 6–19 are ordinary commutative algebra and algebraic geometry in the n constants.

Carrying out either assembly is a set-theoretic computation on constructible sets — note that $\mathbb{C}^n \setminus C_i = (\mathbb{C}^n \setminus V(E_i)) \cup V(h_i)$ is constructible, so all the operands are — and the operation to ask for is the one [14] calls SMPD, *strongly make pairwise disjoint*: given pieces $A_1, \ldots, A_\ell$, it returns pairwise disjoint pieces $B_1, \ldots, B_u$ such that each A_k is a union of some of the B_l. The property passes from the pieces to any union of them, so applying it at line 14 to the pieces making up $\mathbb{C}^n$ and the family $W_1, C_1, \ldots, W_s, C_s$ partitions the constant space and leaves every block B either contained in or disjoint from each W_i and each C_i, so that the assemblies become a selection of blocks:

$$B \subseteq V_\forall \iff B \subseteq W_i \text{ for every } i \text{ with } B \subseteq C_i,$$
$$B \subseteq V_{\exists\forall} \iff B \subseteq W_i \text{ for some } i.$$

Nothing beyond the refinement is computed: the conditions are read off block by block, and the answer is irredundant because the blocks are disjoint, so no containment test is needed. The $\exists\forall$ assembly gets off more lightly still. Its condition mentions neither $\mathbb{C}^n$ nor the cells, so at line 17 it is enough to refine $W_1, \ldots, W_s$ alone — s sets rather than $2s + 1$ — after which every block lies in some W_i by construction and all of them are kept. A component whose cell misses a block imposes nothing on it, and in this formulation that is exact rather than a heuristic — only the components whose cells contain a block are ever consulted for it, which is the computational form of the remark that a component contributes a condition only where its cell lies.

This is what [14] calls the set-theoretic version of the coprime factorization problem; SMPD is its Algorithm 3, built on an algorithm for the difference of two constructible sets presented by *regular systems* — a regular chain together with an inequation, the same shape as the pairs line 11 produces. The constructible sets themselves are available as a type, with the difference and union operations and the weaker MPD (which merely removes duplicated data from a single constructible set, preserving its union), in the `ConstructibleSetTools` module of the `RegularChains` library [15]. There is a second route that is more economical here, being the algorithm of line 1 in its non-differential case: an algebraic Thomas decomposition returns a disjoint decomposition to begin with, and the implementation accompanying [7] represents complements and intersections of solution sets as decompositions into simple systems. Where a *canonical* representation is wanted and not merely an irredundant one, the constructible sets of the parametric Gröbner literature ([16] and its successors) carry one, computed in [17].

Line 19 is a normalization. A block of a refinement is a Boolean combination of the sets refined, and an intersection of irreducible sets need not be irreducible, so the blocks selected at line 15 or 17 need not arrive in the irreducible form the output calls for. Applying to each of their pairs (F, h) the same operation as line 11 restores it, and on a pair already of that form the operation does nothing, $\langle F \rangle$ being prime and not containing h.

Including $\mathbb{C}^n$ among the sets refined at line 14 costs one more argument and makes the consistency of the ansatz a by-product of the decomposition. The components of a Thomas decomposition partition $\mathrm{Sol}(A(c))$ and the cells are their projections, so $\bigcup_i C_i$ is exactly the set of c^* at which $A(c^*)$ has a solution; the ansatz is consistent throughout $\mathbb{C}^n$ if and only if no block of $\mathcal{B}$ lies outside every cell. Inconsistency never shows up as a defective component — every simple system has a solution ([7], Remark 2.3), and DifferentialDecompose discards a branch as soon as it produces an equation $c_=$ with c a non-zero constant, or the inequation $0_{\neq}$ — it shows up as a gap in the covering. For the ansatz $\{\, c\Psi_t - 1 \,\}$, say, the decomposition splits on the initial c, throws away the branch $c = 0$ where the equation becomes $-1_=$, and returns a single component whose cell is $\{\, c \neq 0 \,\}$; the point $c = 0$ lies in no cell.

A gap of this kind costs the $\forall$ assembly nothing. Where $A(c^*)$ is inconsistent, $\sqrt{[A(c^*)]}$ is the unit ideal and contains all of $\mathcal{P}$, so c^* belongs to the membership locus — and a block lying in no cell satisfies the condition of line 15 vacuously and is returned. Under $\mu = \forall$ the algorithm therefore needs no consistency hypothesis at all. The $\exists\forall$ assembly drops such a block instead, so under $\mu = \exists\forall$ the chain $V_\forall \subseteq V_{\exists\forall} \subseteq V_\exists$ does require the ansatz to be consistent, as its right-hand containment requires in any case: the proof in §2.2 carries the hypothesis $1 \notin [A(c^*)]$, and indeed at an inconsistent c^* we have $1 \in [A(c^*), \mathcal{P}]$, so that c^* lies in $V_\forall$ but not in $V_\exists$.

The $\forall$ assembly remains the more expensive of the two. The $\exists\forall$ assembly can be read directly off the output of line 11, using the refinement only to tidy the result, whereas the $\forall$ assembly genuinely needs the complements $\mathbb{C}^n \setminus C_i$, and it is the splitting these force — not the ideal arithmetic, which is elementary — that carries the cost.

The $\exists\forall$ assembly is offered for the reader content with an approximation. It replaces the intersection by a plain union of the W_i, and so never forms a complement — the pieces come out of line 11 ready to be listed. What it computes is not the membership locus but an intermediate one. Because the components of a Thomas decomposition partition the solution set of $A(c^*)$, the loci are distinguished by a pair of quantifiers, one ranging over components and one over the solutions within a component:

locus	components	solutions	all of $\mathcal{P}$ vanishes ...
$V_\forall$	$\forall$	$\forall$	on every solution
$V_{\exists\forall}$	$\exists$	$\forall$	on some entire component
$V_\exists$	$\exists$	$\exists$	at some one solution

The middle row is weaker than the first and stronger than the third, so

$$V_\forall \;\subseteq\; V_{\exists\forall} \;\subseteq\; V_\exists. \tag{41}$$

For the left inclusion, a point of $V_\forall$ lies in some cell — the ansatz is consistent, so it has a solution, and that solution lies on some component — and Corollary 6 then places the point in that component's $\mathcal{V}_i$. For the right inclusion, if $c^* \in \mathcal{V}_i \cap C_i$ then component i carries at least one solution over c^* (Lemma 4) and, by Theorem 5, every element of $\mathcal{P}$ vanishes on all of them; so $A(c^*)$ and $\mathcal{P}$ have a common solution and $1 \notin [A(c^*), \mathcal{P}]$. Note that this is where the system must be taken as a whole: the witnessing solution has to satisfy every P_j at once, and it does, because a single component's solutions are annihilated by all of them simultaneously.

Three cautions attend the $\exists\forall$ assembly. First, both inclusions in (41) can be strict, and the second can be as wide as possible: in Example 2 the ansatz is already simple, so the decomposition has the single component $\{\Psi_{tt} - c\Psi\}$, the reduced PDE $\Psi_t - \Psi$ has coefficients 1 and -1, and $\mathcal{V}_1 = \emptyset$ — whence $V_{\exists\forall} = \emptyset$ against $V_\exists = \mathbb{C}$. This is not a defect of the assembly step but of the projection step that precedes it, which asks each reduced equation to vanish identically and so cannot see a single solution satisfying the system; no rearrangement of the assembly recovers it. Computing $V_\exists$ genuinely requires the system inside the decomposition, which is the computation §2.2 set aside as infeasible. Second, cutting each $\mathcal{V}_i$ down to its cell is not cosmetic: outside C_i the component has no solutions at all, Theorem 5 says nothing, and the right inclusion of (41) fails. Third, $V_{\exists\forall}$ is not intrinsic to the ansatz and the system the way $V_\forall$ and $V_\exists$ are — it depends on which decomposition was computed, since a finer stratification makes "vanishes on an entire component" easier to satisfy.

When every element of $\mathcal{P}$ is homogeneous with respect to Ψ, $V_\exists$ is all of $\mathbb{C}^n$ and the upper bound in (41) says nothing. The useful statement is then obtained by discarding the components on which Ψ itself reduces to zero: restricting the union in line 17 to those i with $\Psi \notin E^{(i)} : (q^{(i)})^\infty$ forces the witnessing solution to be non-trivial, and gives $V_{\exists\forall} \subseteq V_{\exists\setminus\Psi}$. By Proposition 1 the test is one reduction of Ψ modulo $S_i^=$ per component.

Algorithm 3 runs on any ansatz, and with $\mu = \forall$, by Theorem 5 and Corollary 6, what it returns is exactly the membership locus — neither missing valid choices of constants nor admitting spurious ones. The next section proves this.

2.4 Completeness of the Algorithm

Two questions arise. If some particular selection of constants causes the ansatz to solve the system — that is, if it lies in $V_\forall$, the membership locus — will the algorithm always find it? And conversely, is everything the algorithm returns genuinely in $V_\forall$? Theorem 5 answers both at once for a single component of the decomposition, and Corollary 6 assembles the components.

Lemma 4. *Let $(S^=, S^{\neq})$ be a differentially simple system arising as a component of a differential Thomas decomposition of the ansatz, computed with the constant parameters $c = (c_1, \ldots, c_n)$ ranked below all other indeterminates, and let its cell be the projection of its solution set onto $\mathbb{C}^n$, so that every point of the cell carries at least one solution of the component. If c^* lies in the cell, then the specialized system $(S^=(c^*), S^{\neq}(c^*))$ is again differentially simple.*

Proof. Because the constants are ranked below every other indeterminate, the two conditions of algebraic simplicity are quantified over solution sets that contain the whole cell: by [7], Definition 2.2, the initial of each equation is non-zero, and each equation square-free in its leader, at every solution of the subsystem in the strictly lower-ranked variables, and for the equations carrying differential indeterminates that subsystem is the constant part of the component. Both therefore hold at c^*, and (S1) persists.

Specialization evaluates coefficients and introduces no new derivative; since no initial vanishes at c^*, every leader stays in place. Janet's division is computed from the leaders alone, so the multiplicative/non-multiplicative split is unchanged, and each reduction to zero witnessing (S2) remains a valid reduction after its coefficients are evaluated at c^*. Conditions (S3) and (S4) concern which equations and inequations are present and reducible, and no equation or inequation degenerates at a point of the cell. Hence $(S^=(c^*), S^{\neq}(c^*))$ satisfies (S1)–(S4). $\square$

Lemma 4 plays the role that the "specializes well" property of algebraic regular chains plays in the triangular-set literature, where it holds off the zero set of the iterated resultants of the initials and separants ([14], Definition 11 and Proposition 4; the Gröbner-basis precedent is [18]). The difference is where the exceptional locus goes. For a regular differential system it must be carved out afterwards, and the square-free conditions need not survive the specialization at all. A Thomas component instead carries its own conditions on the constants: the parameter values at which the system would degenerate are not in the cell, having been split off into other components of the decomposition. Completeness is recovered by running the algorithm on every component.

Theorem 5. *Let $\mathcal{P} = \{P_1, \ldots, P_t\}$ be a finite system of differential polynomials and let $A(c)$ be the ansatz, a set of differential polynomials whose coef-*

ficients involve constant parameters $c = (c_1, \ldots, c_n) \in \mathbb{C}^n$. Let $(S^=, S^{\neq})$ be one differentially simple component of a differential Thomas decomposition of the ansatz, computed with the constants ranked below all other indeterminates, write $A = \{a_1, \ldots, a_k\}$ for its equations $S^=$, and let $H_A = I_A \cup S_A$ denote the set of initials and separants of A.

Apply Ritt's full reduction to each P_j modulo A, obtaining a remainder r_j and a power product h_j of elements of H_A such that

$$h_j \cdot P_j - r_j \in [A] \qquad (j = 1, \ldots, t) \tag{42}$$

with r_j fully reduced with respect to A. Write $r_j = \sum_\alpha m_{j\alpha} \cdot p_{j\alpha}(c)$, where for each j the $m_{j\alpha}$ are distinct power products in the non-constant indeterminates and the $p_{j\alpha}$ are polynomials in the constants, so that $\operatorname{Coeffs}\big(r_j, \mathbb{Q}[c_1, \ldots, c_n]\big) = \{p_{j1}, \ldots, p_{jN_j}\}$. *Let*

$$\mathcal{V} \;=\; V\Big(\textstyle\bigcup_{j=1}^{t} \operatorname{Coeffs}\big(r_j,\, \mathbb{Q}[c_1, \ldots, c_n]\big)\Big) \;\subseteq\; \mathbb{C}^n$$

be the variety defined by setting all of these constant coefficients to zero.

Let q be the product of the initials and separants of A, and suppose c^* lies in the cell of the component — the projection of its solution set onto $\mathbb{C}^n$. Then

$$c^* \in \mathcal{V} \quad \Longleftrightarrow \quad \mathcal{P} \subseteq [A(c^*)] : q^\infty.$$

In particular, $\mathcal{P} \subseteq \sqrt{[A(c^*)]}$ implies $c^* \in \mathcal{V}$.

Note that each h_j is in general a polynomial in the constants *and* in jet variables ranked below the leaders of A, so the condition that matters is that $h_j(c^*)$ not be the zero polynomial; that is what preserves the degrees of the leaders in $A(c^*)$, and hence what keeps $r_j(c^*)$ fully reduced. Now every factor of h_j is an initial or a separant of an equation of A; by (A2) the initials do not vanish at any solution of the corresponding lower subsystem, and by the consequence of (A3) noted above neither do the separants. Since c^* lies in the cell, some solution a of the component lies above it, and $h_j(a) \neq 0$; hence $h_j(c^*)$ is not identically zero. The h_j need not agree with one another — each reduction clears whatever initials and separants it happens to meet — but the argument is applied to one j at a time and never compares them.

The theorem needs no further hypothesis. The whole of what the proof requires — that reduction to zero decide membership, and that the relevant ideal be radical — is delivered by Proposition 1, whose only hypothesis is that the system be simple. That is exactly what the components of a differential Thomas decomposition are, and Lemma 4 carries the property to the point c^*. This is the structural reason for decomposing the ansatz with a differential Thomas decomposition: its output satisfies the theorem's hypotheses by construction, so nothing remains to be verified component by component.

Because the theorem is an equivalence, it settles both directions at once: the algorithm is complete, in that every valid choice of constants is found, and also sufficient, in that no point of $\mathcal{V}$ is spurious. The ideal characterized is the saturation $[A(c^*)] : q^\infty$, which by Proposition 1 is radical and therefore contains $\sqrt{[A(c^*)]}$; the final clause of the theorem is that containment. The two can differ, so $\mathcal{V}$ may contain constants at which some P_j lies in the saturation without lying in $\sqrt{[A(c^*)]}$ — but the saturation is the vanishing ideal of the component's own solution set, so at such a point the ansatz does solve the system on that component.

Proof. Fix j. Ritt's full reduction algorithm produces h_j and r_j satisfying (42), with r_j fully reduced with respect to A. Evaluating the constants at $c = c^*$ gives $h_j(c^*) \cdot P_j - r_j(c^*) \in [A(c^*)]$. Write $E = [A(c^*)]$. Since c^* lies in the cell, Lemma 4 makes $(S^{=}(c^*), S^{\neq}(c^*))$ a simple differential system, so Proposition 1 applies to it.

Now r_j is fully reduced with respect to A, and as observed above $h_j(c^*)$ is not identically zero, so no leader of A drops in degree upon specialization and $r_j(c^*)$ is fully reduced with respect to $A(c^*)$; it is therefore its own pseudo-remainder. Proposition 1 then gives

$$r_j(c^*) = 0 \quad \Longleftrightarrow \quad r_j(c^*) \in E : q^\infty.$$

Since $h_j(c^*) \cdot P_j - r_j(c^*) \in E \subseteq E : q^\infty$, the right-hand side holds if and only if $h_j(c^*) \cdot P_j \in E : q^\infty$. And h_j is a power product of elements of H_A while q is the product of all of them, so h_j divides some power of q; as $E : q^\infty$ is unchanged by further saturation at q, we get $h_j(c^*) \cdot P_j \in E : q^\infty$ if and only if $P_j \in E : q^\infty$. Chaining,

$$r_j(c^*) = 0 \quad \Longleftrightarrow \quad P_j \in [A(c^*)] : q^\infty.$$

Now the power products $m_{j\alpha}$, for this fixed j, are distinct monomials in a polynomial ring, hence linearly independent over $\mathbb{C}$, so $r_j(c^*) = 0$ if and only if every coefficient $p_{j\alpha}(c^*) = 0$.

Both sides of the equivalence just established are conjunctions over j of statements each involving only that one j: the right-hand condition $\mathcal{P} \subseteq E : q^\infty$ holds if and only if $P_j \in E : q^\infty$ for every j, and $c^* \in \mathcal{V}$ holds if and only if $p_{j\alpha}(c^*) = 0$ for every j and α. Taking the conjunction over $j = 1, \ldots, t$ of the per-equation equivalence therefore gives

$$c^* \in \mathcal{V} \quad \Longleftrightarrow \quad \mathcal{P} \subseteq [A(c^*)] : q^\infty.$$

Note that the ideal $E : q^\infty$ is the same for all j — it depends on the component and on c^*, not on which equation is being reduced — which is what lets the t statements be conjoined without further ado. The last clause follows because $E : q^\infty$ is radical and contains E, hence contains $\sqrt{E}$. $\square$

Theorem 5 concerns a single component. The membership locus of the ansatz as a whole is recovered by intersecting over the decomposition.

Corollary 6. *Let $\mathcal{V}_1, \ldots, \mathcal{V}_s$ be the varieties produced by Theorem 5 from the components $S_1, \ldots, S_s$ of a differential Thomas decomposition of the ansatz. Then, for every $c^* \in \mathbb{C}^n$,*

$$c^* \in V_\forall \quad \Longleftrightarrow \quad c^* \in \mathcal{V}_i \ \text{ for every } i \text{ whose cell contains } c^*.$$

Proof. By Lemma 4 the components whose cells contain c^* specialize to simple differential systems, and their fibres over c^* partition the solution set of $A(c^*)$; the remaining components have no solution over c^* and contribute nothing. The ansatz carries no inequations, so $q = 1$ in [8], Prop. 2.2.72 ([9], Prop. 3.32), and that proposition reads

$$\sqrt{[A(c^*)]} \;=\; \bigcap_i \bigl(E^{(i)} : (q^{(i)})^\infty\bigr),$$

the intersection running over those components. Since $\mathcal{P}$ is contained in an intersection of ideals exactly when it is contained in each of them, $\mathcal{P} \subseteq \sqrt{[A(c^*)]}$ if and only if $\mathcal{P} \subseteq E^{(i)} : (q^{(i)})^\infty$ for every such i; and by Theorem 5 that holds if and only if $c^* \in \mathcal{V}_i$.

No hypothesis of consistency is needed. If no cell contains c^* then $A(c^*)$ has no solution, the intersection above is empty and so equals the whole ring, and both sides of the equivalence hold: $\mathcal{P}$ lies in $\sqrt{[A(c^*)]} = (1)$, and the condition on the right is vacuous. $\square$

The two quantifiers the corollary combines — over the components of the decomposition and over the equations of the system — are independent, and both are conjunctions, so the order in which they are discharged is immaterial. The algorithm discharges the inner one first, pooling the coefficients of all t remainders into a single J_i before the component's minimal primes are computed, which is why the cost of the assembly is unchanged by the size of the system.

Determining which cells contain a given c^* requires no computation. Because the constants form a block below every non-constant indeterminate and the component is triangular, its equations and inequations sort by leader: those whose leader is a non-constant indeterminate involve that indeterminate and lower-ranked quantities, while the rest involve only the constants and the independent variables. Of the latter, those lying in $\mathbb{Q}[c_1, \ldots, c_n]$ are the cell. Indeed, by [7], Remark 2.3, every solution of the subsystem in the strictly lower-ranked variables extends to a solution of the next level of a simple system, and conversely every solution restricts; so, inductively, the projection of $\mathrm{Sol}(S^=, S^{\neq})$ onto $\mathbb{C}^n$ is exactly the solution set of the constant-only part of $(S^=, S^{\neq})$. Deciding whether c^* lies in the cell is therefore a matter of substituting c^* into finitely many polynomials in the constants and checking that the equations vanish and the inequations do not. The cell is thus quasi-affine — cut out by equations *and* inequations — and cells belonging to different components may overlap, even though the components themselves are disjoint, since a case distinction made at the level of a derivative refines the fibre without refining the base.

All of this rests on the constants being differential indeterminates in their own right, ranked as a block, rather than elements of the coefficient ring. Buried in the coefficients they would be inert: no decomposition would ever split on them, and there would be no cells to read off. It is the same choice that lets the method solve for a constant — an energy eigenvalue, say — rather than be handed it.

This is a different operation from the projection step of §2.3, and the two should not be confused. There we ask each reduced equation to vanish *identically*, and so set the coefficient of every power product to zero; here the equations of the component are constraints cutting out its solution set, not identities holding on it, and their coefficients are not required to vanish at all.

So the per-component test decides membership in the *saturated* ideal of that component, while intersecting over the decomposition decides membership in the *radical* differential ideal of the ansatz — which is the membership locus as defined in §2.2. Note that this is an intersection of loci in the space of constants, not a union: a c^* at which even one equation of $\mathcal{P}$ fails on even one component is not in $V_\forall$, since the ansatz there has solutions that do not satisfy the system.

3 Example: Hydrogen Schrödinger Equation

We seek to demonstrate the operation of the algorithm on the following Schrödinger equation for hydrogen [3]:

$$-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi \tag{43}$$

where ∇ is the Laplacian, Ψ is a complex-valued wavefunction defined over real 3-dimensional space, and r is the distance to the origin. We choose to work in Cartesian coordinates, so $r^2 = x^2 + y^2 + z^2$, and we use Hartree atomic units to render the equation dimensionless. This is the case $t = 1$ of §2: the system $\mathcal{P}$ has the single element obtained by clearing the denominator in (43), and we write P for it throughout this section. The Navier–Stokes example of §4 exercises the general case.

We work in the following differential ring:

$$R \;=\; \mathbb{Q}[x,y,z]\{\,\Psi,\Psi',\Psi'',v,r,\; E,a_0,a_1,b_0,b_1,c_0,c_1,v_1,v_2,v_3,v_4\,\} \tag{44}$$

with $\Delta = \{\delta_x, \delta_y, \delta_z\}$. To elaborate,

- the independent variables x, y, z lie in the coefficient ring, where $\delta_i x_j = \delta_{ij}$;
- $E, a_0, \ldots, v_4$ are the eleven constants, sent to zero by every derivation. E is a constant of the PDE rather than of the ansatz, but the algorithm makes no distinction: all eleven are ranked together in the lowest block, and the membership locus is a subset of $\mathbb{C}^{11}$ having E among its coordinates. This is what allows the eigenvalue to be solved for rather than supplied;
- $\Psi, \Psi', \Psi'', v, r$ are the five non-constant differential indeterminates. They will ultimately be mapped into a function space, so it is often convenient to think of them as functions; algebraically, all that is asserted of them here is that they are not constants. Note that Ψ' and Ψ'' are separate indeterminates, not Δ-derivatives of Ψ — they are tied to Ψ by the chain-rule equations of the ansatz, and r likewise carries $r^2 - x^2 - y^2 - z^2$ in the system rather than as a quotient of the ring;
- the ranking is specified below.

We'll use the first ansatz from section 1, a second order ODE with first degree coefficients and a first degree independent variable. (Figure 2).

To equations (17), (18) and (19) we append $r^2 - x^2 - y^2 - z^2$ to handle the algebraic extension used to construct the PDE, and obtain the following system of differential polynomials:

$$\Psi_x - \Psi' v_x \qquad \Psi_y - \Psi' v_y \qquad \Psi_z - \Psi' v_z \tag{45a,b,c}$$
$$\Psi'_x - \Psi'' v_x \qquad \Psi'_y - \Psi'' v_y \qquad \Psi'_z - \Psi'' v_z \tag{45d,e,f}$$
$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi \tag{45g}$$
$$v - (v_1 x + v_2 y + v_3 z + v_4 r) \tag{45h}$$
$$r^2 - (x^2 + y^2 + z^2) \tag{45i}$$

To use Ritt's reduction algorithm, we must now specify a ranking. While the algorithm's correct operation does not depend on any particular ordering, different rankings can produce dramatically different performance timings. We use this orderly ranking:

$$\{\Psi'' > \Psi' > \Psi > v > r\} \gg \{E, a_0, \ldots, v_4\} \gg \{x, y, z\} \tag{46}$$

in the block form of Algorithm 3: the five non-constant indeterminates, ordered among themselves as shown, above the eleven constants, above the independent variables.

A differential Thomas decomposition of ansatz (45) with ranking (46) produces 29 cells, and the algorithm of section 2 then computes the following four solution varieties for PDE (43):

$$\big(v_4 c_0 - b_0, E c_0 - v_4 c_1, -v_4^2 c_1 + E b_0, b_1, 2a_1 - b_0, a_0, v_3, v_2, v_1\big) \tag{47a}$$
$$\big(v_4 c_0 - b_0, v_1^2 + v_2^2 + v_3^2 - v_4^2, c_1, b_1, a_1 - b_0, a_0, E\big) \tag{47b}$$
$$(-b_1 c_0 + b_0 c_1, v_4 c_1 - b_1, v_4 c_0 - b_0, a_1, a_0, v_3, v_2, v_1, 2E + 1) \tag{47c}$$
$$\big(c_0^2 + 4 b_0 c_1, 4 v_4 c_1 + c_0, v_4 c_0 - b_0, 2b_1 + c_0, a_1, a_0, v_3, v_2, v_1, 8E + 1\big) \tag{47d}$$

In brief,

(47a) corresponds to the classical radial solutions,

(47b) is a solution in parabolic coordinates,

(47c) is a first-order solution for the 1s shell, and

(47d) is a first-order solution for the 2s shell.

As an example, let's analyze ideal (47a). $a_0 = 0$, and we can always multiply our homogenous differential polynomial by a constant without affecting our result, so we can set $a_1 = 1$. Likewise, we can multiply our variable v by a constant and that will only change our coefficients by constants, and $v_1 = v_2 = v_3 = 0$, so we can normalize by setting $v_4 = 1$. This immediately implies that $b_0 = c_0 = 2$ and simplifies ideal (47b) to:

$$(v_3, v_2, v_1, v_4 - 1, c_0 - 2, 2E - c_1, b_1, 2 - b_0, a_1 - 1, a_0) \tag{48}$$

Setting these polynomials to zero and substituting into (17) and (19), we obtain:

$$\begin{gathered} v = r \\ v\Psi'' + 2\Psi' + 2(1 + Ev)\Psi = 0 \end{gathered} \tag{49}$$

This is the classical radial equation obtained by seperation of variables ([3] §18a). We use spherical coordinates, set $\Psi = R(r)P(\theta)F(\psi)$, and obtain the above equation for $R(r)$, though it is more commonly written in this form:

$$\frac{1}{R}\frac{d}{dr}\left[r^2\frac{dR}{dr}\right] + 2(Er^2 + r) = l(l+1) \tag{50}$$

where l is the orbital quantum number, which can be zero. Set $l = 0$, $v = r$ and $\Psi = R$ and expand out the derivative to obtain (49).

Ideal (47b) is the most unusual of the four. $a_0 = 0$ and we can set $a_1 = 1$, by the same logic as above. If we limit our v_i coefficients to be real, all of their squares must be positive, and since $v_1^2 + v_2^2 + v_3^3 - v_4^2 = 0$, v_4 must be non-zero. So we can normalize by setting $v_4 = 1$. Substituting in these values, we find ideal generators that imply $b_0 = 1$ and simplify (47b) to this ideal:

$$\left(v_1^2 + v_2^2 + v_3^2 - 1, v_4 - 1, c_0 - 1, c_1, b_1, 1 - b_0, a_1 - 1, a_0, E\right) \tag{51}$$

This ideal corresponds to the following system of equations:

$$
\begin{aligned}
E &= 0 \\
a_0 = 0 &\qquad a_1 = 1 \\
b_0 = -1 &\qquad b_1 = 0 \\
c_0 = -1 &\qquad c_1 = 0 \\
v_1^2 + v_2^2 + v_3^2 = 1 &\qquad v_4 = 1
\end{aligned}
\tag{52}
$$

Substituting these values back into our ansatz, we conclude that $\Psi(v)$ is a solution of (43) under these conditions:

$$
\begin{gathered}
v\Psi'' + \Psi' + \Psi = 0 \\
v = v_1 x + v_2 y + v_3 z + r \\
v_1^2 + v_2^2 + v_3^2 = 1
\end{gathered}
\tag{53}
$$

The expression $v_1 x + v_2 y + v_3 z$ is easily identified as a dot product between the coordinate (x, y, z) and the unit vector (v_1, v_2, v_3) (remembering that $v_1^2 + v_2^2 + v_3^2 = 1$). The direction of this vector is arbitrary, so we can orient the x-axis in this direction and set $(v_1, v_2, v_3) = (1, 0, 0)$ without loss of generality.

We now have to solve a second order ODE:

$$v\Psi''(v) + \Psi'(v) + \Psi(v) = 0 \tag{54}$$

Wolfram Mathematica [19] can now analyze this equation and determine that it is solved by the Bessel function (55).

```
In[1]:= DSolve[x*y''[x] + y'[x] + y[x] == 0, y[x], x]

Out[1]= {{y[x] -> BesselJ[0, 2 Sqrt[x]] C[1] + 2 BesselY[0, 2 Sqrt[x]] C[2]}}
```

leading to the following exact solution to (43), with $E = 0$:

$$\Psi = J_0(2\sqrt{x + r}) \tag{55}$$

where x, y, z are Cartesian coordinates, $r = \sqrt{x^2 + y^2 + z^2}$, $E = 0$, and J_0 is the ordinary Bessel function J_0.

(43) admits a separation of variables in parabolic coordinates ([20] §37):

$$\xi = r + z \qquad \eta = r - z \qquad \psi = \tan^{-1} \frac{y}{x} \tag{56}$$

Solution (55) is in this parabolic form.

The result can be easily verified using Wolfram Mathematica [19], as follows:

```
In[1]:= Psif := Psi[x, y, z]

In[2]:= r[x_, y_, z_] = Sqrt[x^2 + y^2 + z^2]

                2    2    2
Out[2]= Sqrt[x  + y  + z ]

In[3]:= eqn = - 1 / 2 * Laplacian[Psif, {x, y, z}] - 1 / r[x, y, z] * Psif == 0

                                    (0,0,2)              (0,2,0)              (2,0,0)
            Psi[x, y, z]       -Psi        [x, y, z] - Psi        [x, y, z] - Psi        [x, y, z]
Out[3]= -(------------------) + ---------------------------------------------------------------- == 0
                  2    2    2                                 2
          Sqrt[x  + y  + z ]

In[4]:= sol[x_, y_, z_] := BesselJ[0, 2 * Sqrt[x + r[x, y, z]]]

In[5]:= FullSimplify[eqn /. Psi -> ({x, y, z} |-> sol[x, y, z])]

Out[5]= True
```

The remaining ideals are first order ODEs and can be analyzed in a similar fashion, yielding the following solutions:

$$(-b_1c_0 + b_0c_1, v_4c_1 - b_1, v_4c_0 - b_0, a_1, a_0, v_3, v_2, v_1, 2E + 1)$$

$$E = -\tfrac{1}{2} \qquad v = r \qquad \Psi' + \Psi = 0$$

$$\Psi = e^{-r}$$

$$\left(c_0^2 + 4b_0c_1, 4v_4c_1 + c_0, v_4c_0 - b_0, 2b_1 + c_0, a_1, a_0, v_3, v_2, v_1, 8E + 1\right)$$

$$E = -\tfrac{1}{8} \qquad v = r \qquad (1 - \tfrac{1}{2}v)\Psi' + (1 - \tfrac{1}{4}v)\Psi = 0$$

$$\Psi = (2 - r)e^{-r/2}$$

In short, the algorithm presented in section 2 finds solutions expressed in both first and second order differential equations, in both spherical and cylindrical coordinates.

4 Example: Incompressible Navier–Stokes

The hydrogen example ran the simplest ansatz of Section 1 against a single linear PDE. We now run the *last* one — the nonlinear, second-degree ansatz of Figure 8, whose four top-level polynomials u, v, w, p were introduced precisely to serve as the dependent variables of a system — against the incompressible Navier–Stokes equations.

Where the derivation below lands is worth having in view from the start. It lands on the *normalized truncation* of that ansatz drawn in Figure 9: the ODE has been truncated to its linear part, the six second-degree jet monomials dropped, and the symmetries of the problem have been used up to cut the twenty-eight constants of Figure 8 (with the linear backgrounds of u, v, w removed) down to nine. Rotations of (x, y, z) put the covector $\mathbf{s}$ along x and the amplitude $\mathbf{a}$ in the x–y plane, so that $s_2 = s_3 = 0$ and $w \equiv 0$; the three scalings the ansatz admits — multiplying the ODE by a constant, scaling Ψ, scaling s — are pinned by $g_0 = 1$, $v_5 = 1$ and $s_1 = 1$; and the mean pressure gradient is set to zero, $p_1 = p_2 = p_3 = 0$. What is left is

$$\begin{gathered}
\Psi_c = \Psi' s_c, \qquad \Psi'_c = \Psi'' s_c \qquad (c = x, y, z, t) \\
(1 + g_1 s)\,\Psi'' + (h_0 + h_1 s)\,\Psi' + (i_0 + i_1 s)\,\Psi = 0 \\
s = x + s_4 t \\
u = u_5 \Psi, \qquad v = \Psi, \qquad w = 0, \qquad p = p_4 t + p_5 \Psi
\end{gathered} \tag{57}$$

in the nine constants s_4; g_1, h_0, h_1, i_0, i_1; u_5; p_4, p_5. Figure 9 draws (57) in the notation of Figure 1: there are nine white boxes, one for each of those constants. The three coefficients of the ODE are first degree polynomials in s and their lines are labelled accordingly, the leading one differing from the others only in having had its constant term normalized to 1. The remaining lines carry no degree bound, because the polynomials they belong to are confined to part of their ring — s to two of the four coordinates, u, v, w, p to Ψ and t — and are written out instead; v and w, fixed outright, carry no box at all.

None of these reductions is free, and each is earned below rather than assumed: the truncation to a linear ODE is the cell $a_0 = a_1 = 0$ that the degree mismatch of §4.4 forces on any viscous solution, and the vanishing of the mean pressure gradient is a conclusion of §4.5, not a hypothesis. Two cautions attach to the picture. The rotation used here differs from the one used by hand in §4.5, which sends $\mathbf{s}$ along z and $\mathbf{a}$ along x; the two describe the same solution in different coordinates. And truncating the ODE, unlike the rotation and the scalings, is not a change of chart but a change of family: Figure 9 is a smaller ansatz, and what it finds is not automatically a result about Figure 8 until the second-degree block is separately shown to be inert, which is what §4.4 does.

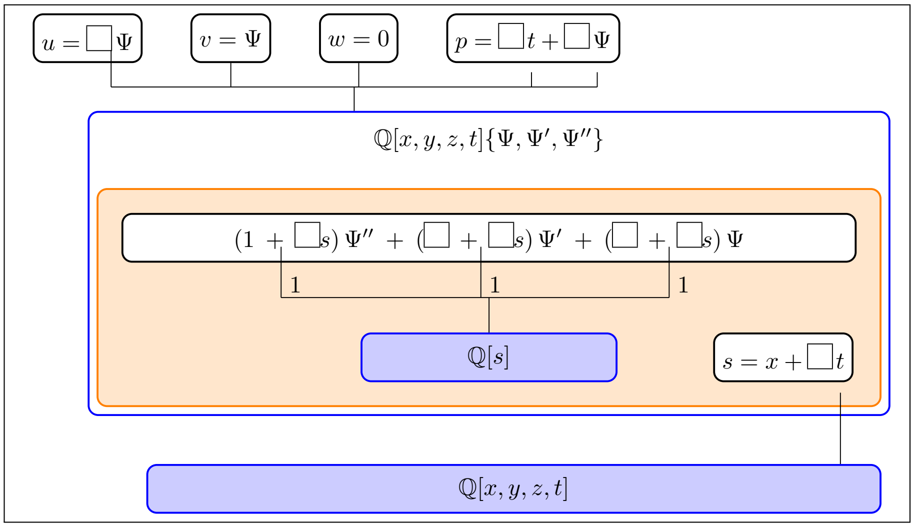

Figure 9: The normalized truncation of Figure 8 used in the computation: a linear ODE in place of the second-degree one, the rotation and scaling freedoms spent, and no mean pressure gradient. Polynomials that a normalization has restricted are written out rather than drawn as a single box, and the nine white boxes are the nine free constants.

The purpose of the example is not to produce a new solution of Navier–Stokes. It is that this particular pairing of ansatz and PDE reduces in closed form, by hand, and that the closed form says something sharp about which solutions the ansatz can reach and, more interestingly, why it cannot reach the others. Two obstructions appear, and neither is visible from the ansatz diagram alone: the nonlinearity of the equations is annihilated by their own incompressibility condition, and the viscous term is blocked by a mismatch of degrees between the ansatz's ODE and the PDE.

4.1 The equations

For constant density ρ and dynamic viscosity μ (kinematic viscosity $\nu = \mu/\rho$), the incompressible Navier–Stokes equations are

$$u_x + v_y + w_z = 0 \tag{58a}$$
$$\rho\,(u_t + uu_x + vu_y + wu_z) + p_x - \mu\,(u_{xx} + u_{yy} + u_{zz}) = 0 \tag{58b}$$
$$\rho\,(v_t + uv_x + vv_y + wv_z) + p_y - \mu\,(v_{xx} + v_{yy} + v_{zz}) = 0 \tag{58c}$$
$$\rho\,(w_t + uw_x + vw_y + ww_z) + p_z - \mu\,(w_{xx} + w_{yy} + w_{zz}) = 0 \tag{58d}$$

Unlike the hydrogen equation (43), these require no clearing of denominators. Once u, v, w, p are replaced by the polynomials of (25) they are already differential polynomials in

$$R \;=\; \mathbb{Q}[x, y, z, t]\{\,\Psi, \Psi', \Psi'', s, u, v, w, p,\; \rho, \mu, a_0, \ldots, i_1, s_1, \ldots, s_4, u_1, \ldots, p_5\,\} \tag{59}$$

with $\Delta = \{\delta_x, \delta_y, \delta_z, \delta_t\}$. As with E in the hydrogen example, nothing compels us to treat ρ and μ as given; they are ranked in the lowest block alongside the constants of the ansatz, and the membership locus is free to constrain them. It will do exactly that below.

4.2 Reduction

Because s is linear in the coordinates, its derivatives $s_x = s_1$, $s_y = s_2$, $s_z = s_3$ and $s_t = s_4$ are constants, and the chain rules of (25) collapse the first derivatives of each field to

$$u_x = u_1 + u_5 s_1 \Psi' \quad u_y = u_2 + u_5 s_2 \Psi' \quad u_z = u_3 + u_5 s_3 \Psi' \quad u_t = u_4 + u_5 s_4 \Psi' \tag{60}$$

and likewise for v, w and p with their own coefficients. Applying the chain rules a second time, every Laplacian in (58) collapses to a single term,

$$u_{xx} + u_{yy} + u_{zz} \;=\; u_5\,\sigma\,\Psi'', \qquad \text{where} \quad \sigma \;=\; s_1^2 + s_2^2 + s_3^2. \tag{61}$$

It is convenient to name two vectors of constants,

$$\mathbf{s} = (s_1, s_2, s_3), \qquad \mathbf{a} = (u_5, v_5, w_5),$$

$\mathbf{s}$ being the spatial part of the covector that defines the ODE's independent variable and $\mathbf{a}$ the amplitude with which the ODE element enters the velocity field, and to write U, V, W, P for the parts of u, v, w, p that are linear in the coordinates, so that $u = U + u_5\Psi$, $p = P + p_5\Psi$ and $\bar{\mathbf{v}} = \mathbf{L} + \mathbf{a}\Psi$ with $\mathbf{L} = (U, V, W)$.

Substituting into (58) and collecting on the jets gives the reduced system

$$(u_1 + v_2 + w_3) \; + \; (\mathbf{a}\cdot\mathbf{s})\,\Psi' \tag{62a}$$

$$\begin{aligned} &\rho\big[\partial_t\mathbf{L} + (\mathbf{L}\cdot\nabla)\mathbf{L}\big] + \nabla P \; + \; \rho\left[(\mathbf{a}\cdot\nabla)\mathbf{L}\right]\Psi \\ &\quad + \big[\rho s_4\,\mathbf{a} + \rho(\mathbf{L}\cdot\mathbf{s})\,\mathbf{a} + p_5\,\mathbf{s}\big]\,\Psi' \; + \; \rho\,(\mathbf{a}\cdot\mathbf{s})\,\mathbf{a}\,\Psi\Psi' \; - \; \mu\sigma\,\mathbf{a}\,\Psi'' \end{aligned} \tag{62b}$$

the three momentum residuals being the three components of (62b). Its x component, for instance, has jet-free part $\rho u_4 + p_1 + \rho(u_1U + u_2V + u_3W)$ and Ψ coefficient $\rho(u_1u_5 + u_2v_5 + u_3w_5)$; note that the u_5 there is a component of $\mathbf{a}$ and stays put in the y and z components, where the gradient row (u_1, u_2, u_3) becomes (v_1, v_2, v_3) and (w_1, w_2, w_3). Every jet monomial appearing anywhere in (62) is drawn from the five-element set $\{1,\ \Psi,\ \Psi',\ \Psi\Psi',\ \Psi''\}$. Nothing else occurs — in particular no Ψ''^2, no $\Psi''\Psi'$ and no Ψ'^2.

4.3 The convective nonlinearity

Navier–Stokes is nonlinear only through the convective derivative $(\bar{\mathbf{v}}\cdot\nabla)\bar{\mathbf{v}}$. Under this ansatz the velocity field is a fixed direction $\mathbf{a}$ modulated by a single scalar profile Ψ of a single scalar argument s, superposed on a background $\mathbf{L}$ linear in the coordinates. Expanding the x component of the convective derivative in those terms separates it into four pieces:

$$\begin{aligned} (\bar{\mathbf{v}}\cdot\nabla)u \;=\; &\underbrace{(\mathbf{L}\cdot\nabla U)}_{\text{background–background}} \;+\; \underbrace{(\mathbf{a}\cdot\nabla U)\,\Psi}_{\text{wave–background}} \\ &+\; \underbrace{u_5(\mathbf{L}\cdot\mathbf{s})\,\Psi'}_{\text{background–wave}} \;+\; \underbrace{u_5(\mathbf{a}\cdot\mathbf{s})\,\Psi\Psi'}_{\text{wave–wave}} \end{aligned} \tag{63}$$

Only the last piece is quadratic in the jets, and it is the only quadratic jet monomial in the entire system. Its coefficient carries the factor $\mathbf{a}\cdot\mathbf{s}$ — which is precisely the coefficient of Ψ' in the continuity residual (62a).

The consequence is immediate, but worth stating carefully, because it is a property of the ansatz and not an approximation. The decomposition splits on $\mathbf{a}\cdot\mathbf{s}$.

On the cells where continuity holds with a non-degenerate Ψ' we have $\mathbf{a} \cdot \mathbf{s} = 0$, and then the wave–wave term of (63) vanishes identically — in all three momentum equations at once, since their coefficients are $\rho u_5(\mathbf{a} \cdot \mathbf{s})$, $\rho v_5(\mathbf{a} \cdot \mathbf{s})$ and $\rho w_5(\mathbf{a} \cdot \mathbf{s})$. What is left is linear in Ψ, Ψ' and Ψ''.

The condition $\mathbf{a} \cdot \mathbf{s} = 0$ has a clean reading: the amplitude of the wave is orthogonal to the direction in which the wave varies. The field is transverse — it is a shear wave, and the profile varies only along a direction in which the velocity does not point. A fluid dynamicist would recognize the conclusion, since unidirectional flow has no self-advection, but here it is not assumed. It is one of the branches the decomposition returns.

Two qualifications keep this from being stronger than it is. First, only the wave's self-interaction is destroyed. The background–wave term $u_5(\mathbf{L} \cdot \mathbf{s})\Psi'$ survives; it is linear in the jets, but its coefficient is linear in x, y, z, t, so it contributes several distinct equations to the constant locus, and the example is not a linear problem in disguise. Second, the jet-free row of (63) contributes $(\mathbf{L} \cdot \nabla)\mathbf{L} = 0$ up to the pressure's linear part, the familiar requirement that the background be one of the linear strain-and-rotation flows.

What this says about the ansatz is that its second-degree monomials are not being asked for by this PDE. Figure 8 was drawn with Ψ''^2, $\Psi''\Psi'$, $\Psi''\Psi$, Ψ'^2, $\Psi'\Psi$ and Ψ^2 all available, and the natural reading of the figure is that the ansatz's nonlinearity is there to meet Navier–Stokes' nonlinearity. It is not. The residual's only quadratic term is $\Psi\Psi'$, and incompressibility removes it. A second-degree ODE buys a wider class of profile functions Ψ, which is a different thing — and as the next subsection shows, one of those six monomials actively costs us something.

4.4 How the viscous term is blocked

The Laplacian is the only source of Ψ'' in (62). It enters each momentum residual exactly once, through (61), and it enters linearly. Each residual is therefore of degree *one* in Ψ''. The ansatz's ODE, on the other hand, is of degree *two* in Ψ'', on account of its $(a_0 + a_1 s)\Psi''^2$ term.

Ψ'' is the leader of both under any ranking that places the jets above the constants and orders $\Psi'' \gg \Psi' \gg \Psi$. Ritt's reduction (§2.1) can lower the degree of a differential polynomial r in the leader of A only when $\deg_{\Psi''} r \geq \deg_{\Psi''} A$, and here $1 < 2$. Nor does full reduction rescue the situation: the proper derivatives of A introduce Ψ''', and the residuals contain no Ψ''' at all. Each momentum residual is thus *already fully reduced* modulo the ansatz. Pseudo-division returns it unchanged, and the ODE — the entire content of the figure's orange box — contributes nothing whatever to the reduction.

The algorithm therefore passes the residual to the projection step untouched, and the projection sets each of its coefficients to zero separately. The Ψ'' coefficients give

$$\mu u_5 \sigma \;=\; \mu v_5 \sigma \;=\; \mu w_5 \sigma \;=\; 0. \tag{64}$$

If $\mu \neq 0$, then either $\sigma = 0$ or $\mathbf{a} = 0$. Over the reals $\sigma = s_1^2 + s_2^2 + s_3^2 = 0$ forces $\mathbf{s} = 0$, so that $s = s_4 t$ and the profile has no spatial dependence whatever, while $\mathbf{a} = 0$ removes Ψ from the velocity field altogether and leaves only the linear background. On this branch, then, Navier–Stokes admits nothing but inviscid or spatially degenerate solutions.

It is worth being precise about the culprit, because it is easy to read (64) as a statement about fluids. It is not. The obstruction is that a PDE of degree one in a leader cannot be reduced by an ansatz of degree two in that same leader. The higher degree does not add solutions; it makes the ansatz inert.

The remedy is already inside the algorithm. The Thomas decomposition of the ansatz splits on the initial of its ODE, so the locus $a_0 = a_1 = 0$ is one of the cells the decomposition produces, not something we have to impose by hand. On that cell the ODE reads

$$\begin{gathered}\big[(b_0 + b_1 s)\Psi' + (c_0 + c_1 s)\Psi + (g_0 + g_1 s)\big]\Psi'' \\ + (d_0 + d_1 s)\Psi'^2 + (e_0 + e_1 s)\Psi'\Psi + (f_0 + f_1 s)\Psi^2 \\ + (h_0 + h_1 s)\Psi' + (i_0 + i_1 s)\Psi \;=\; 0\end{gathered} \tag{65}$$

which is of degree one in Ψ'', and pseudo-division by it now does something. Note how little has to be given up. Of the six second-degree monomials only Ψ''^2 is switched off; the other five survive, and the two that involve the leader, $\Psi''\Psi'$ and $\Psi''\Psi$, are positively useful, in that they make the initial of (65) a non-constant differential polynomial on which the decomposition splits again, so that the degenerate loci are reported as cells of their own rather than silently excluded.

Stated generally, this is a criterion for the selection of an ansatz (§6): the degree of the ansatz's defining polynomial in a given leader should not exceed the degree in which that leader occurs in the PDE, or the ansatz cannot act on the PDE at all.

4.5 The traveling-wave shear layer

Work on the cell $a_0 = a_1 = 0$, and take the linear background to vanish, $\mathbf{L} = 0$, so that $u = u_5\Psi$, $v = v_5\Psi$, $w = w_5\Psi$ and $p = p_1x + p_2y + p_3z + p_4t + p_5\Psi$. The continuity residual (62a) then reduces to $\mathbf{a}\cdot\mathbf{s} = 0$ alone. Since $\mathbf{a}$ and $\mathbf{s}$ are both unknowns of the problem we may rotate coordinates freely; choose $\mathbf{s}$ along z and $\mathbf{a}$ along x, so that

$$s = s_3z + s_4t, \qquad \mathbf{a} = (u_5, 0, 0), \qquad \sigma = s_3^2, \qquad \mathbf{a}\cdot\mathbf{s} = 0.$$

The y and z momentum residuals then have $v \equiv w \equiv 0$ and collapse to $p_2 = 0$ and $p_3 + p_5s_3\Psi' = 0$, whence $p_3 = 0$ and $p_5 = 0$: the pressure carries no part of the ODE element. The x residual is what remains,

$$p_1 \;+\; \rho u_5 s_4\,\Psi' \;-\; \mu u_5 s_3^2\,\Psi'' \;=\; 0. \tag{66}$$

Its jet-free coefficient gives $p_1 = 0$, no mean pressure gradient — the specialization to $u_4 = 0$ of the general jet-free condition $p_1 = -\rho u_4$, in which a uniformly accelerating background balances a pressure gradient. What is left,

$$\mu s_3^2\,\Psi'' - \rho s_4\,\Psi' \;=\; 0, \tag{67}$$

is not a further constraint but the ansatz's own ODE. Matching it against (65) fixes $b_* = c_* = d_* = e_* = f_* = i_* = 0$ and $g_1 = h_1 = 0$, with

$$\frac{h_0}{g_0} \;=\; -\,\frac{\rho s_4}{\mu s_3^2}.$$

The membership locus is therefore nonempty. The reduction does not annihilate the ansatz's coefficients; it pins them.

Integrating (67) gives the solution

$$\Psi' \;\propto\; \exp\!\left(\frac{\rho s_4}{\mu s_3^2}\,s\right), \qquad u = u_5\,\Psi(s_3z + s_4t), \qquad v = w = 0, \tag{68}$$

with the pressure a function of t alone, which is to say an additive gauge. Writing $c = -s_4/s_3$ for the speed at which the profile translates and $\zeta = z - ct$ for the co-moving coordinate, so that $s = s_3\zeta$, (68) takes the form

$$u(\zeta) \;=\; A + B\, e^{-c\zeta/\nu}, \qquad \nu = \mu/\rho, \tag{69}$$

an exponential shear layer of thickness ν/c propagating rigidly along z at speed c. This is the traveling-wave member of the Rayleigh–Stokes family of viscous shear solutions: a front separating two streams, held at a fixed thickness by the balance between viscous diffusion, which thickens it, and the translation, which sweeps vorticity through it. In profile it is the same exponential as the asymptotic suction boundary layer. That (69) solves (58) exactly is immediate — the convective term vanishes identically because the flow is unidirectional with $u_x = 0$, which is (63) with $\mathbf{L} = 0$ and $\mathbf{a} \cdot \mathbf{s} = 0$.

It is instructive that the *original* Rayleigh problem is not in this family. Stokes' first problem, the impulsively started plate, has the self-similar solution $u = U\,\mathrm{erfc}\big(z/2\sqrt{\nu t}\big)$, a function of $z/\sqrt{t}$. This ansatz cannot express it, and the reason is visible in the diagram rather than in the analysis: the ODE's independent variable s is drawn as a first-degree polynomial in the base ring $\mathbb{Q}[x, y, z, t]$, and $z/\sqrt{t}$ is not a polynomial in x, y, z, t at all. Reaching it would mean adjoining an algebraic element τ with $\tau^2 = t$ and defining the independent variable by the cleared relation $\tau s - z$ — the algebraic-extension mechanism of Figure 4, applied to the base rather than to the ODE's coefficients.

4.6 What the example shows

The machinery of §2 runs unchanged on a nonlinear system of four PDEs in four dependent variables, and its output is informative even where the answer is largely negative. The solutions it returns are exactly the parallel shear flows, and the two reasons it returns nothing else are structural.

Removing the second obstruction costs one monomial: drop Ψ''^2, or equally well work on the cell $a_0 = a_1 = 0$ that the decomposition hands us anyway. Removing the first is a deeper change. As long as u, v and w are affine in one common ODE element Ψ, the velocity is a single profile times a fixed direction, and $(\bar{\mathbf{v}} \cdot \nabla)\bar{\mathbf{v}}$ cannot survive incompressibility. Letting the components carry *different* ODE elements — several extensions adjoined in parallel, in the sense of §5 — is what an ansatz would need in order to reach a flow whose nonlinearity is active.

5 Generalization of the Algorithm

The algorithm of §2 already accepts a system of any size — that is what the loop of lines 4–7 is for — so the generalizations that remain are on the other side of the computation, in the ansatz. A collection of sample ansätze was presented in Section 1; this section sketches how much further they can be pushed.

Multiple ODE extensions can be introduced, either in parallel, allowing the ODE solutions to be combined together into a polynomial that solves the PDE, or in series (Figure 3), allowing either the independent variable of one ODE to be formed from the solutions to a second ODE. Extensions can also be introduced in the coefficient ring of the ODE (Figures 6 and 7)

In addition to ODE extensions, algebraic extensions can be introduced either explicitly or implicitly. Figure 4 shows an explicit second degree algebraic extension. Introducing higher degree algebraic extensions is also possible. Figure 5, a somewhat simpler variant on the same idea, shows an ansatz that introduces a square root, then allows the ODE's independent variable to be selected from the resulting algebraic extension.

Parameterized polynomials of any (finite) degree can be used in lieu of the linear polynomials. Figure 5 illustrates the use of second degree polynomials. Polynomials can be further restricted. Perhaps we have some theoretical consideration which suggests that the independent variable should be symmetric.

Nonlinear ODEs can be introduced, as shown in Figure 6.

Nonlinear ODEs permit algebraic extensions to be introduced implicitly. Consider that for a given algebraic extension:

$$f_n(x)y^n + \cdots + f_1(x)y + f_0(x) \qquad f_i(x) \in \mathbb{C}[x] \tag{70}$$

we can form the derivative with respect to x as follows:

$$y' = \frac{f_n'(x)y^n + \cdots + f_1'(x)y + f_0'(x)}{nf_n(x)y^{n-1} + \cdots + f_1(x)} \tag{71}$$

This is a non-linear ODE that is first order, n^{th} degree in y.

Note that if we limit our ansatz to linear ODEs, then any algebraic extensions need to be introduced explicitly; i.e, a linear ODE will not match algebraic extensions.

We can introduce inhomogenous ODEs very similarly to the nonlinear case by introducing a zeroth degree coefficient into the differential equation.

Do we need to restrict the coefficients of the ODE to a ring generated by the independent variable? We could allow coefficients in the ODE to involve indeterminates other than v, and there are two major cases to consider. The first case is when we can partition the indeterminates into two sets, one used to express the independent variable, and one used to express the coefficients of the ODE. This yields a *parameterized ODE*, one whose coefficients vary according to one set of independent variables (the parameters), and whose independent variable depended on a different set of them. The parameters can be treated as constants for the purpose of solving the ODE. I don't know of any way to enforce this requirement other than having multiple ansätze, each with a different partitioning of the indeterminates. One large ansatz could be constructed that would enforce nothing, but would still allow for ODE solutions of the form described next.

What if the dependencies of the ODE coefficients and the ODE variable can't be partitioned into two different sets of variables?

Then we're going to get an ODE something like this (two dimensional, for illustrative purposes):

$$f(u,v)\frac{d^2\Psi}{dv^2} + g(u,v)\frac{d\Psi}{dv} + h(u,v)\Psi = 0 \tag{72}$$

the point being that we can't separate the v dependency out of the coefficients.

I'm not aware of any theory to handle an ODE of the form (72), but one might exist, or be developed. Would it be useful? Perhaps. (72) is still simpler than a PDE because it only involves one derivative. The algorithm described in this paper could test for the existence of such solutions. I'm somewhat skeptical of their utility.

6 Selection of an Ansatz

The most obvious question remains: how to select an ansatz?

First, can we build any such ansatz to express all of the solutions of the PDE? This seems unlikely, as the solutions of a PDE would include solutions for all possible boundary conditions.

For example, the simple heat equation $u_t = u_{xx}$ admits an infinity of solutions of the form

$$u(x,t) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-x_0)^2}{4t}} \qquad \forall x_0 \in \mathbb{R} \tag{73}$$

No single exponential extension can be constructed to express all of these solutions. In fact, a different extension is required for each real value of x_0, yielding a uncountable infinite basis.

Furthermore, these are just the solutions described in a series expansion; we expect there to be many others. The essence of the problem of ansatz selection is that we expect there to be solutions all over the place. Many ansätze will yield solutions.

So, while the PDE problem is superficially similar to the problem of symbolic integration, which asks if a finite tower of extensions can be used to express an integral, the nature of the problem is quite different because in the integration case we can express all of its solutions in a single extension. In fact, given one solution to the integration problem, we know that all of its solutions are related by adding a constant C, which implies that a field containing one solution contains all solutions. In the symbolic integration case, we expect to find a single field extension that contains all solutions, while in the PDE case we expect this to be impossible.

What about differential Galois theory?

Picard-Vessiot / differential Galois theory has been of great utility in analyzing ordinary differential equations. An n^{th} order linear ODE, will have at most an n-dimensional solution space. After selecting an arbitrary basis, any solution can be represented by an n-tuple of complex constants. Any automorphism that permutes these solutions can be represented by an $n \times n$ matrix, so differential Galois theory can be reduced to the analysis of $GL_n(\mathbb{C})$, which has facilitated the work of Kovacic, Singer, and Bronstein.

No such easy reduction is available for PDEs. A PDE with no boundary conditions is likely to have a solution space that is not only infinite, but infinite dimensional, and probably not even with a countably infinite basis. In such an environment, differential Galois groups can not be represented as subgroups of GL_n, so the theory of algebraic groups is therefore unlikely to be much help.

Kovacic developed an algorithm [21] to handle second order ODEs not by developing an algorithm to determine the differential Galois group of a given equation, but rather by using first a change of variables to restrict the Galois group to SL_2, then categorizing all possible algebraic subgroups of SL_2 and developing an algorithm to handle each case. Taking a similar approach for PDEs would require categorizing all possible linear mappings between function spaces, which does not seem feasible.

In short, the differential algebra techniques used to analyze integrals and linear ODEs do not seem amenable to handling PDEs, and no theory is available at this time to guide the selection of an ansatz. The ansatz used for the hydrogen example was an educated guess. It was designed to be general enough to find the existing classical solution, and so to be used for testing the software. The appearance of (47b) was not anticipated, as the solutions was previously unknown, and serves to illustrate the mysterious nature of these ansätze.

7 Software

The calculations in section 3 were done with SageMath [22], a computer algeba system that integrates dozens of open source mathematical programs and libraries into a unified, Python-based framework, though stock Sage is generally lacking in differential algebra tools. Therefore, the author used an AI coding assistent (Claude Opus) to prepare two additional Sage packages.

The first is a Sage interface to François Boulier's BLAD libraries, which is used for many fundamental differential algebra operations, such as representing differential rings, polynomials and rankings, as well as Ritt reduction. The package is available here:

`https://github.com/BrentBaccala/sage-differential-polynomial`

Second, for the differential Thomas decomposition, the author prepared a Sage port of Maple's `DifferentialThomas` library, available here:

`https://github.com/BrentBaccala/DifferentialThomas-sage`

The script that used these packages to perform the calculations in this paper is available here:

`https://github.com/BrentBaccala/Papers/NewMethod`

`thomas-ansatz-solve.sage` is verified to work with Sage 10.6 and the two libraries described above.

Computing a differential Thomas decomposition of (45) with ranking (46) takes about seven minutes on an AMD Ryzen 5 processor and produces 29 cells. Performing the differential reduction / projection / irreducible decomposition pipeline on each of those cells takes another half hour and computes the following four solution varieties:

8 Future Work

The difficulties of selecting an ansatz have already be touched upon. One possible approach to ansatz selection would be to impose boundary conditions (or the like) prior to any kind of analysis with differential algebra. In the case of quantum mechanics, imposing $H^2(\Omega)$, where Ω is the state space of whatever physical system we're analyzing, is an obvious restriction. Some kind of theoretical analysis, exploring what restrictions must be placed on ansätze to force the functions into $H^2(\Omega)$, might yield useful techniques to better select ansätze.

What relationship is there between the constant ideals and the function space?

The utility of this algorithm is currently limited by our Gröbner basis software. For example, I have formed a system of 1570 equations with 1,296,960 terms to test whether helium can be solved using ansatz (22). Various attempts to compute a decomposition of this system into its minimal associated primes exhaust the memory on a computer with 96 GB RAM, but the software lacks the ability to fall back effectively on mass storage when RAM is exhausted, nor can it use memory spread out over a cluster.

9 Conclusion

The algorithm presented above can be used to check any PDE to see if any of its solutions can be expressed using an ODE structured according to a specific ansatz, defined by differential polynomials and parameterized by a finite number of constants. This technique is complementary to separation of variables, where we check a PDE to see if any of its solutions can be expressed as a product of factors, each depending on only a single variable, with no restrictions to differential polynomials. It is likewise complementary to the *similarity-reduction* methods — the classical Lie symmetry method [23], the Clarkson–Kruskal direct method [24, 25], and the nonclassical (conditional) symmetries of Bluman and Cole [26] — which also reduce a PDE to an ODE without assuming a separable product form, but which are organized around the (classical or conditional) symmetries of the equation. Our reductions are instead organized around the differential-algebraic structure of the ansatz; they do not presuppose any symmetry of the PDE, and they carry the completeness guarantee of Theorem 5.

The name of the software repository (`helium`) informs a primary goal of my research: to find a closed form expression for helium's ground state. In that repository, you will find several years of attempts in that direction, of which `joca.sage` is the most refined. Although "solving helium" is widely regarded as impossible, I know of no proof that helium's ground state wavefunction can not be formed from a finite number of linear homogenous ODEs. Lacking any good theory to inform the selection of an ansatz, I have turned to educated guesswork.

We can explore other important PDEs, such as the Navier-Stokes equation. As with helium, we can work incrementally towards it using smaller ansätze. Ansatz (25) is a step in the direction of analyzing Navier-Stokes. $\bar{\mathbf{v}} = (u, v, w)$ is meant to be a velocity field and p is the pressure, all varying as functions of position (x, y, z) and time t.

We can solve differential equations with Gröbner bases, and this technique, given enough compute resources, allows us to check for simple ODE-based solutions, but we're primarily limited by inefficiencies of our software.

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